BCSE CLASS X
PHYSICS

Exam Notes with Solved Papers 2023, 2024 & 2025

10 chapters · summary tables · formulas · diagrams · every question of the three papers solved

How to use these notes

Each chapter has (1) a **must-know table** of the key ideas, (2) a **formula table**, (3) **diagrams** you may be asked to draw or label, and (4) **solved questions** from the 2023, 2024 and 2025 papers, grouped by topic. Calculations show **formula → substitution → answer with unit**, the way marks are awarded.

Please note

Some questions depend on a diagram in the paper (for example the generator, transformers and the stand-and-clamp pictures). I read these diagrams carefully, and wherever a diagram was unclear I say so and give the method as well. Use **g = 10 m/s²** unless the question gives another value (the 2024 MCQ options are based on g = 10).

Contents

All chapters in order, with page numbers

Chapter	What it covers / where it was asked	Page
Paper Pattern & Question Map	What the examiner asks — and how often	**3**
Chapter 1: Moments, Couples, Centre of Gravity, Equilibrium & Stability	Appears in Section A and Section B every year	**5**
Chapter 2: Gravitation, Free Fall & Terminal Velocity	Section A and Section B in all three years	**11**
Chapter 3: Work, Energy, Power, Efficiency & Energy Resources	Calculations and "design / justify" questions in all three years	**14**
Chapter 4: Pressure, Pascal's Law & Hydraulic Machines	Every year: at least one hydraulic calculation and several pressure MCQs	**18**
Chapter 5: Heat, Specific Heat Capacity & Calorimetry	A specific-heat calculation appears every year	**21**
Chapter 6: Current Electricity — Ohm's Law, Resistance & Heating Effect	V–I graphs and a current / power calculation appear every year	**25**
Chapter 7: Electromagnetism, Generators & Transformers	A transformer calculation and a Fleming-rule question appear every year	**28**
Chapter 8: Waves, the Electromagnetic Spectrum & Communication	The most question-rich chapter: 10–12 items in every paper	**32**
Chapter 9: Radioactivity & Nuclear Energy	A decay equation and an alpha/beta/gamma question appear every year	**38**
Chapter 10: Space Exploration & the Universe	An "explain / compare / justify" question comes from this chapter every year	**41**
Formula Sheet & Last-Minute Revision	Physics — everything on two pages	**44**

Paper Pattern & Question Map

What the examiner asks — and how often

1. Structure of the paper (same in 2023, 2024, 2025)

Section	Questions	Marks
A (compulsory)	**Q1**: (a) 25 multiple-choice questions [25] • (b) fill in the blanks [5] • (c) true / false [5] • (d) match Column A with Column B [5]. (In 2023 the order of the last three parts is fill-in, true/false, match.)	40
B (any six)	**Q2 to Q9** — eight questions, attempt any **six**. Each question has parts (a) to (e or f) worth 1, 2 or 3 marks: short explanations, calculations, diagram-based and "justify / design" questions.	60

> **Smart choice in Section B**
>
> Questions are mixed-topic (each of Q2–Q9 mixes mechanics, heat, electricity, waves and nuclear parts). Choose the six questions in which you can answer the most parts, and always attempt the calculations — their marks are the easiest to secure.

2. Topic-wise map (where each topic appeared)

Chapter	2023	2024	2025
1 Moments, couples, centre of gravity, equilibrium & stability	MCQ i, ii, xiii, xiv; Q2a, Q3a, Q4a, Q6a, Q7a, Q9a, b	MCQ iii, viii, ix, x; Fill i; Q2c, Q3e, Q4d, Q8c, Q9c	MCQ i, ii, xiv, xv; Match iv; Q2a, Q3a, Q5b, Q6a, d, Q7a, Q9a
2 Gravitation, free fall & terminal velocity	MCQ iii, xxiii; T/F i; Q3f, Q5a, Q7b, Q8a	Fill ii; T/F iv; Match ii; Q3a, Q4a, Q6e, Q7e, Q9b	MCQ iii, xvi; Fill ii; Q4a, Q7b, c, Q8a
3 Work, energy, power & energy efficiency	MCQ iv, xv, xix; T/F ii; Match i; Q3b, Q4b, Q8b, e, Q9c	MCQ ii, xv, xxi; T/F i, iii; Q2a, Q4b, c, Q5b, Q7c, Q9d	MCQ iv, xvii; Fill iii; T/F i; Match i; Q2b, Q5a, Q7f, Q9b
4 Pressure & Pascal's law	MCQ v, vi; Fill ii; Match ii; Q4c, Q5c, Q6b, Q7c, Q8c	MCQ i, xi, xii, xxii; Q2b, Q3c, Q6a, b	MCQ v, vi, xviii, xix; Q3b, Q5c, Q6c, Q8b, Q9c
5 Heat & calorimetry	MCQ vii, xvi, xxv; Fill iii; Match iii; Q3c, Q5d, Q6c, Q8d	MCQ v; Fill iii; T/F v; Match iii, v; Q3d, Q4e, Q5a, Q6d, Q8d	MCQ vii, xx; Fill iv; T/F ii; Match ii; Q2c, Q4b, e, Q8c, Q9e
6 Current electricity	MCQ xvii; Fill iv; T/F iii; Q3d, Q7d	MCQ xix; Fill v; Match i; Q2d, e	MCQ viii, xxi; T/F iii; Q3c, Q6b, Q9d
7 Electromagnetism, generators & transformers	MCQ viii, xviii; Q2b, Q4d, Q5e, Q9d	MCQ xx, xxv; Match iv; Q6c, Q7a, Q8e	MCQ ix, xxii; Q4c, Q5d, Q8e
8 Waves, EM spectrum & communication	MCQ ix, x, xx, xxi; Fill v; Match v; Q2c, d, Q3e, Q5f, Q6d, Q7e, Q9e, f	MCQ iv, vi, xiii, xvi, xviii, xxiii; Q3b, Q5e, Q7b, d, Q8a, Q9e	MCQ x, xi, xxiii, xxiv; Fill v; T/F iv; Match iii; Q2d, Q3d, Q6e, Q7e, Q8d, Q9f
9 Radioactivity & nuclear energy	MCQ xi, xxii; T/F iv; Q2e, Q4e, Q5b, Q6e, Q8f	MCQ vii, xiv; Fill iv; Q5d, Q8b, Q9a	MCQ xii, xxv; T/F v; Q4d, Q5e, Q6f
10 Space exploration & the universe	MCQ xii, xxiv; T/F v; Q2f, Q4f, Q7f	MCQ xvii, xxiv; T/F ii; Q5c	MCQ xiii; Fill i; Match v; Q2e, Q3e

Repeated every year

Moments and stability • hydraulic machine / Pascal's law • specific heat calculation • transformer (turns ratio) • alpha, beta, gamma properties and decay equation • EM wave uses • free fall / terminal velocity • a "design" or "justify" question on energy efficiency or space exploration.

How to read the Ref column: "2023 MCQ-ii" = 2023 Section A, multiple-choice question ii • "2023 Fill-i" / "T/F-i" / "Match-i" = the fill-in, true/false and matching parts • "2023 Q2a" = Section B, Question 2, part (a).

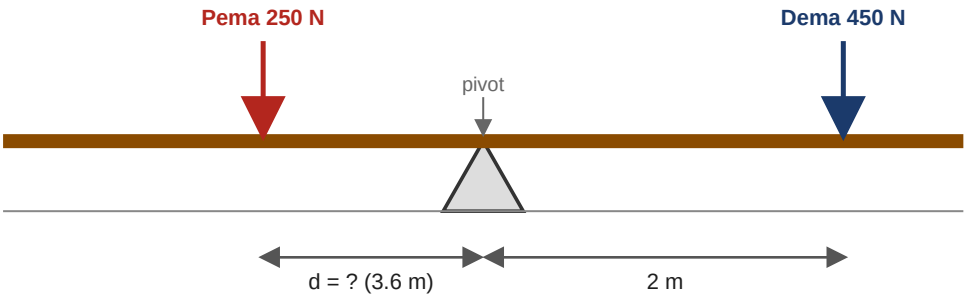
Chapter 1: Moments, Couples, Centre of Gravity, Equilibrium & Stability

Appears in Section A and Section B every year

1.1 Must-know summary

Term	Meaning / rule
Moment of a force (torque)	The turning effect of a force about a pivot. **Moment = force × perpendicular distance from the pivot**. Unit: **N m**. Bigger force or longer arm → bigger moment.
Principle of moments	When a body is in equilibrium, the **sum of the clockwise moments = sum of the anticlockwise moments** about any point.
Couple	Two **equal, opposite, parallel** forces acting at different points (not along the same line). Net force = **0**, but they give a **pure turning effect**. **Moment of couple = F × perpendicular distance between the forces**. Examples: turning a steering wheel, a key, a tap, a bottle cap, a screwdriver, pedalling a bicycle.
Moment of a force vs couple	Force: *one* force about a pivot (opening a door by its handle, using a spanner). Couple: *two* forces, no pivot needed, net force zero.
Centre of gravity (C.G.)	The single point where the **whole weight** of a body appears to act. Uniform square or rectangle: at the centre (a 6 cm × 6 cm square: (3, 3)).
Resultant force & equilibrant	The **resultant** is the single force equal to the net effect of all forces. The **equilibrant** is the force that balances the unbalanced forces (equal and opposite to the resultant).
Equilibrium of a body	Two conditions: (1) **net force = 0** (no change in motion); (2) **net moment = 0** (no turning).
Types of equilibrium	**Stable**: when slightly displaced the C.G. rises and the body returns (bottle standing on its base). **Unstable**: C.G. falls and the body moves further away (pencil on its tip). **Neutral**: C.G. stays at the same height; it stays in the new position (bottle on its side, a ball).
Stability depends on	(1) **Position of the C.G.** — the lower, the more stable. (2) **Area of the base** — the larger, the more stable. A body topples when the vertical line through its C.G. falls **outside** its base.
Everyday uses	Lower C.G.: loaded lorries kept low, racing cars. Wide base: Buddha Dordenma statue, tripod, pyramids. Spreading arms or using a walking stick widens the support and keeps the line of gravity within the base.

Formula	Use
Moment = F × d	wrench, door, see-saw, lever, torque
$F_1d_1 = F_2d_2$ (clockwise = anticlockwise)	balanced beams / levers
Resultant = difference (opposite directions) or sum (same direction)	forces on a body

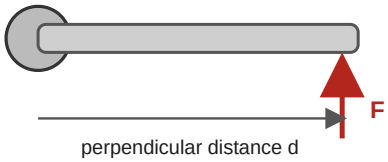


Principle of moments: clockwise moment = anticlockwise moment

450 N x 2 m = 250 N x d => d = 900 / 250 = 3.6 m

Fig 1.1 Principle of moments: the see-saw of 2023 Q2(a)

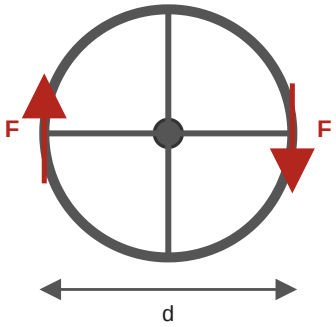
Moment of a force



Moment = F x d (about the nut)

one force: also tends to move the body

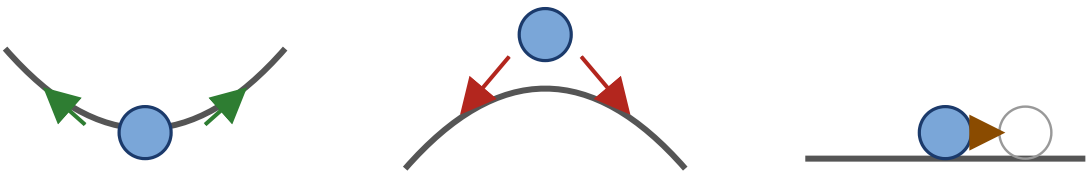
Moment of a couple



Moment of couple = F x d (d = distance between the forces

two equal, opposite, parallel forces: net force = 0, pure rotation

Fig 1.2 Moment of a force vs moment of a couple



STABLE equilibrium

returns to original position

C.G. rises when displaced

(bottle standing upright)

UNSTABLE equilibrium

falls further away

C.G. falls when displaced

(pencil on its tip)

NEUTRAL equilibrium

stays in the new position

C.G. stays at the same height

(bottle lying on its side, ball)

Fig 1.3 Three types of equilibrium

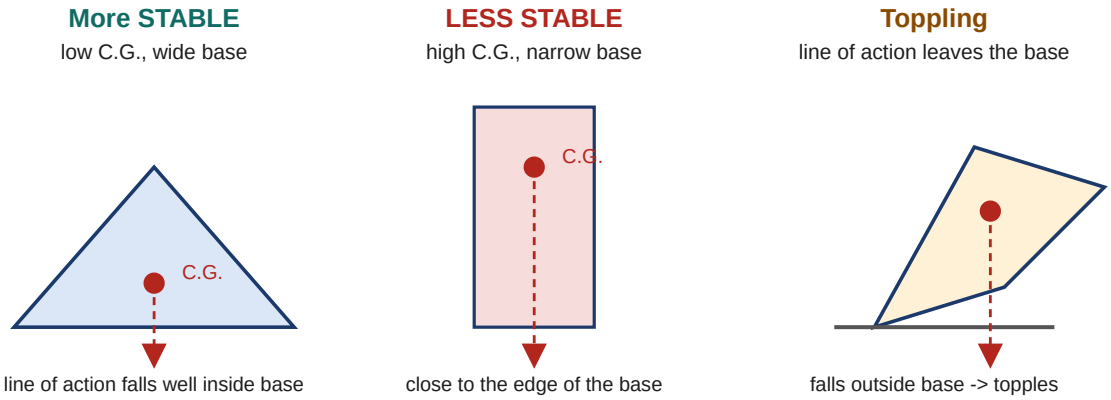


Fig 1.4 Stability depends on the C.G. height and the base area

1.2 Solved questions — Section A (MCQ, fill-in, matching)

Ref	Question	Ans	Why
2023 MCQ-i	The single point where the total weight of a body acts is the…	**C**	centre of gravity.
2023 MCQ-ii	All are examples of a couple EXCEPT… (door, paddling a bicycle, bottle cap, lock with key)	**A**	Opening a door by the handle uses a **single** force about the hinges (moment of a force). The others use two equal and opposite forces (couple).
2023 MCQ-xiii	The item with the **least** stability: ball, umbrella, hour glass, flower vase	**B**	The umbrella standing on its tip has the smallest base and a high C.G. (the ball is only in neutral equilibrium; the hour glass and vase have wide bases).
2023 MCQ-xiv	Who lifts the box with the least force F using a plank and pivot?	**A (Pema)**	Least effort needs the pivot **closest to the load** (longest effort arm, shortest load arm). In Pema's picture the pivot is next to the box.
2024 MCQ-iii	C.G. of a 6 cm × 6 cm square section	**C (3, 3)**	Uniform square: C.G. at the centre, 3 cm from each side.
2024 MCQ-viii	Stability is improved when… (I base larger, II C.G. raised, III base smaller, IV C.G. lowered)	**D (IV & I)**	A larger base and a lowered C.G. increase stability.
2024 MCQ-ix	Where should the force be applied on the wrench for maximum moment?	**D (d)**	The nut is gripped at the left end, so the point **farthest from the nut** (end d) gives the longest arm.
2024 MCQ-x	Moment of a 30 N force on a handle 0.5 m long	**B**	M = F × d = 30 × 0.5 = **15 N m**.
2025 MCQ-i	Which of four cones is the most stable?	**D (cone 4)**	Cone 4 has the widest base and its centre of gravity K is the lowest.
2025 MCQ-ii	A body acted on by 2050 N (east-pointing), 2000 N (west-pointing), 1800 N and 1800 N (north and south): which way does it move?	**A (East)**	North and south forces cancel. East–west: 2050 − 2000 = 50 N towards the east.
2025 MCQ-xiv	How does spreading the arms help a rope-walker?	**A**	It widens the body's mass distribution/support so the **line of gravity stays within the body's base** (the rope), preventing a fall.

2025 MCQ-xv	Torque at the base: 20 N applied at 2 m	**B**	Torque = F × d = 20 × 2 = **40 N m**.

Ref	Question	Answer and working
2023 Fill-i	Force that balances the unbalanced forces is called…	**equilibrant**
2024 Fill-i	The net force from a couple is always…	**zero** (the two forces are equal and opposite).
2025 Match-iv	Product of force and perpendicular distance between two forces	**b. Torque** (moment of a couple).

1.3 Solved questions — Section B

Ref	Question	Answer and working
2023 Q2(a)	A 6 m plank is pivoted 2 m from Dema's end. Dema (450 N) sits on that end. Where must Pema (250 N) sit to balance?	Clockwise moment = anticlockwise moment: 450 N × 2 m = 250 N × d ⇒ d = 900 / 250 = **3.6 m from the pivot** on the other side (0.4 m from the end of the plank, which is 4 m long on that side). (Fig 1.1)
2023 Q3(a)	Complete the table for the two bottle pictures (upright bottle; bottle lying on its side).	**Upright bottle: stable equilibrium** — if tilted slightly its C.G. rises, and it returns to its original position when released. **Bottle on its side: neutral equilibrium** — if rolled, its C.G. stays at the same height and it stays in the new position. (Unstable equilibrium: a body that, when displaced, has its C.G. lowered and moves further away.)
2023 Q4(a)	State the principle of moments.	When a body is in equilibrium, the **sum of the clockwise moments about a point equals the sum of the anticlockwise moments about the same point**.
2023 Q6(a)	Which truck has lesser stability? Why?	**Truck B**: its tall, covered load raises the centre of gravity and the base is narrow compared with its height, so a small tilt can bring the line of gravity outside the base. Truck A has a low flat body and a low C.G.
2023 Q7(a)(i)	How can the mechanic increase the torque of the wheel wrench?	Use a **longer wrench (or add a pipe to the handle)** and push at the far end, perpendicular to the handle: torque = F × d increases with d.
2023 Q7(a)(ii)	How do larger steering wheels help drivers of heavy vehicles?	A large wheel gives a **larger distance between the two hand forces**, so the moment of the couple is larger for the same force: the driver needs less effort to turn the heavy vehicle.
2023 Q9(a)	Which stand-and-clamp set-up is most stable?	**B**. The flask is clamped **low** (lowest centre of gravity) and lies **directly above the base**. A and C hold the flask high; D is low but overhangs the edge of the base.
2023 Q9(b)(i)	Resultant force on the plane (20 000 N forward, 7000 N backward, 3000 N up, 3000 N down) and on the car (2000 N forward, 2000 N backward).	**Plane:** vertical forces cancel; horizontal: 20 000 − 7000 = **13 000 N forward**. **Car:** 2000 − 2000 = **0 N**.
2023 Q9(b)(ii)	Which body is at equilibrium?	The **car**: the resultant force is zero (it is at rest or moving at constant velocity). The plane has a resultant of 13 000 N, so it accelerates.
2024 Q2(c)	ONE difference between the moment of a force and the moment of a couple, with an example of each.	**Moment of a force:** a *single* force turning a body about a pivot; M = F × perpendicular distance from the pivot; e.g. pushing a door at its handle. **Moment of a couple:** *two* equal, opposite parallel forces with net force zero (pure rotation); M = F × distance between the forces; e.g. turning a steering wheel / opening a bottle cap.

2024 Q3(e)	A 1 m ruler pivoted at its midpoint carries 2 N at 30 cm from the left end. Where (X from the right end) must a 10 N force act?	Distance of 2 N from the pivot = 50 − 30 = 20 cm. Distance of 10 N from the pivot = 50 − X. Moments: 2 × 20 = 10 × (50 − X) ⇒ 40 = 500 − 10X ⇒ **X = 46 cm** (0.46 m from the right end).
2024 Q4(d)	Compare the stability of raft designs A and B (table).	**Design A: more stable**; C.G. **low**; barrels in one flat layer with a wide base and the students kneeling, so the line of gravity stays inside the base even when it tilts. **Design B: less stable**; C.G. **high**; barrels stacked in two layers with narrower base and the students standing, so a small tilt brings the line of gravity outside the base.
2024 Q8(c)	Flowchart to find the C.G. of an irregular piece of cardboard.	See Fig 1.5: make three small holes near the edge → hang from a pin through hole 1 → hang a plumb-line from the pin and draw it → repeat with holes 2 and 3 → the lines cross at the C.G. → check: the cardboard balances on a pencil tip there.
2024 Q9(c)	A tray held horizontally by fingers and thumb has three forces U, V and W. Give TWO relationships for equilibrium.	(1) **Net force = 0:** the upward force equals the sum of the downward forces (V = U + W). (2) **Net moment = 0:** by the principle of moments about a point (e.g. the finger-support), the clockwise moments equal the anticlockwise moments (e.g. U × its distance = W × its distance).
2025 Q2(a)	Why does the 54 m Buddha Dordenma statue have a wide base?	A wide base **lowers the C.G. relative to the base and keeps the line of gravity well within the base**, so the statue is very stable against wind and earthquakes; it also spreads the huge weight over a large area, reducing the pressure on the hill.
2025 Q3(a)	Can the man move the rock using the lever? (F = 800 N at 2 m, W = 1800 N at 1 m)	Moment of the effort = 800 × 2 = 1600 N m. Moment of the load = 1800 × 1 = 1800 N m. 1600 < 1800, so **no, he cannot move the rock**. He needs F ≥ 1800 × 1 / 2 = 900 N, or a longer lever.
2025 Q5(b)	Waiter's tray: find the force of the thumb. (fingers act 0.10 m from the left end; the 12 N weight acts 0.25 m from the left end)	Take moments about the finger support: thumb force × 0.10 = 12 × (0.25 − 0.10) = 12 × 0.15 ⇒ force = 1.8 / 0.10 = **18 N** (assuming the distances as drawn).
2025 Q6(a)	Toppled bucket rolled and settled: which equilibrium?	**Neutral equilibrium**: lying on its side, the bucket can roll without its C.G. rising or falling, so it stays wherever it stops.
2025 Q6(d)	Why is turning a screwdriver an application of a couple?	The fingers/hand apply **two equal, opposite parallel forces on opposite sides of the handle**; the net force is zero but they produce a turning moment (rotation without translation).
2025 Q7(a)	How does a walking stick help an old person's stability?	It **increases the area of the base of support** (an extra point of contact) so the line of gravity falls inside the base; it also shares the body's weight.
2025 Q9(a)(i)	X, Y, Z = 25 N each: which has the least moment about O?	**Y**: its line of action passes closest to O, so the perpendicular distance is the smallest (moment = F × d with equal F).
2025 Q9(a)(ii)	Which has the greatest moment about O?	**Z**: its line of action is farthest from O, giving the longest perpendicular distance.

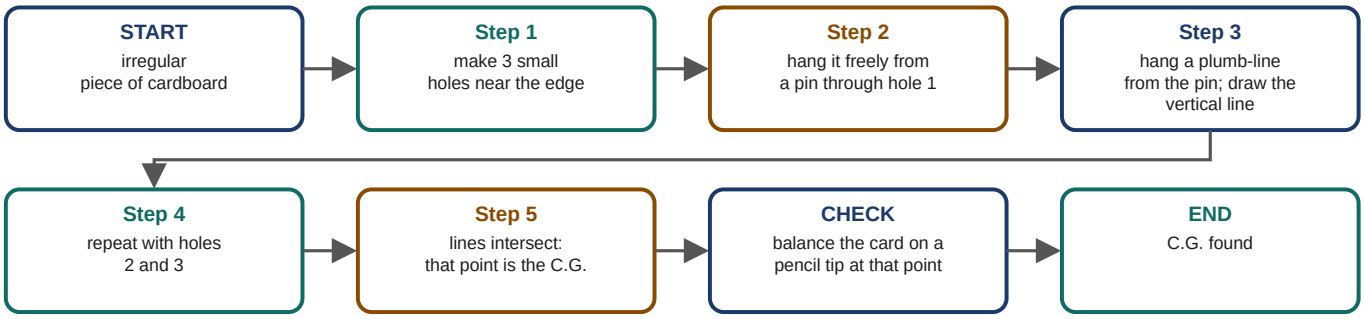
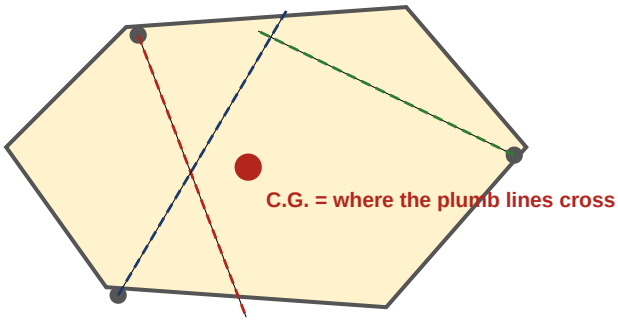


Fig 1.5 Flow-chart: finding the centre of gravity of an irregular lamina (2024 Q8c)

Finding the centre of gravity of an irregular lamina



hang from hole 1, 2, 3 in turn; draw the plumb-line each time

Fig 1.6 The three plumb-lines meet at the centre of gravity

Chapter 2: Gravitation, Free Fall & Terminal Velocity

Section A and Section B in all three years

2.1 Must-know summary

Term	Meaning / rule
Law of universal gravitation	Every body attracts every other body with a force that is **directly proportional to the product of their masses** and **inversely proportional to the square of the distance** between their centres: **$F = G\,m_1m_2 / r^2$**. $G = 6.67 \times 10^{-11}$ N m² kg^{-2}.
Two factors	(1) the **masses** of the two bodies, (2) the **distance** between them. Double both masses → F becomes **4×**. Double the distance → F becomes **¼**.
Weight and g	W = mg. g ≈ 9.8 m/s² (≈ 10) on Earth and about 1.6 m/s² on the Moon. g does **not** depend on the mass of the falling body.
Free fall	Motion under the action of **gravity alone** (air resistance negligible, e.g. a coin on the Moon, a hammer and a feather in the Apollo 15 demonstration). All bodies fall with the **same constant acceleration g**; velocity increases uniformly: v = g t (from rest).
Air resistance (drag)	Opposes the fall; increases with speed and surface area. A sheet of paper falls more slowly than a paper ball of the same mass because its larger area gives a larger drag.
Terminal velocity	As speed grows, drag grows until **drag = weight (driving force)**. Net force = 0 → **acceleration = 0** → the body falls with a **constant velocity**. Before it: net force and acceleration decrease but are not zero.
Microgravity	In space an astronaut's pen cannot write because the ink is not pulled to the nib by gravity.

Formula	Use
$F = G\,m_1m_2 / r^2$	gravitational force; to find a mass: $m_2 = F\,r^2 / (G\,m_1)$
$W = m\,g \quad v = g\,t \quad h = \tfrac{1}{2}\,g\,t^2 \quad v^2 = 2\,g\,h$	free fall (from rest)

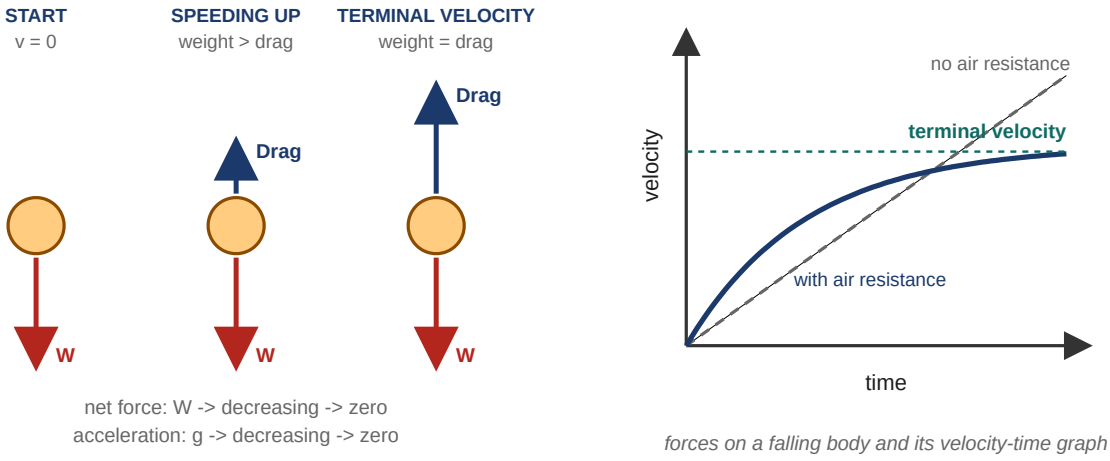


Fig 2.1 Forces on a falling body (left) and its velocity-time graph (right)

2.2 Solved questions — Section A

Ref	Question	Ans	Why
2023 MCQ-iii	Object X falls through air of density 1.37 (P), 1.27 (Q), 1.36 (R), 1.75 (S) kg/m³ from the same height. In which case does it reach the ground first?	**B (Q)**	Air resistance is smallest in the least dense air (1.27 kg/m³), so the object accelerates most and arrives first.
2023 MCQ-xxiii	If the masses of two objects are doubled, the gravitational force…	**B**	$F \propto m_1 m_2 \rightarrow 2 \times 2 = 4$: it **increases by a factor of four**.
2025 MCQ-iii	"Free fall" is the condition in which…	**B**	the motion of a body is due to gravity alone, when air resistance is negligible.
2025 MCQ-xvi	When do falling objects attain terminal velocity?	**B**	When the driving force (weight) equals the drag force: net force zero, velocity constant.

Ref	Question	Answer and working
2023 T/F-i	The acceleration of a falling body at terminal velocity is zero.	**TRUE** — weight and drag balance, net force is zero.
2024 Fill-ii	Acceleration due to gravity for a falling object is unaffected by its…	**mass**.
2024 T/F-iv	A falling coin on the Moon is an example of free fall.	**TRUE** — the Moon has no air, so only gravity acts.
2024 Match-ii	Acceleration due to gravity is zero (Column A)	**c. terminal velocity** — at terminal velocity the net force and hence the acceleration of the falling body is zero.
2025 Fill-ii	A body that falls at a constant velocity will be falling under the sole influence of…	**gravity** (expected answer). *Note:* strictly, a body at constant (terminal) velocity has balanced forces; "sole influence of gravity" describes free fall with constant *acceleration*.

2.3 Solved questions — Section B

Ref	Question	Answer and working
2023 Q3(f)	Why can't astronauts write with a normal pen in space?	A normal pen needs **gravity to make the ink flow** to the nib. In space (microgravity) the ink does not flow down, so the pen does not write.
2023 Q5(a)	A cat falling from a building is not injured. Why?	(1) Its small mass and large body area give a **low terminal velocity** (air resistance soon balances its weight), so it does not fall faster and faster. (2) It spreads its body and lands on its flexible legs, bending them to **increase the time of stopping** and reduce the force of impact.
2023 Q7(b)	An object falls from the same height on the Moon and on the Earth. Compare the velocities and give a reason.	The object reaches the **greater velocity on the Earth** because g is larger on the Earth (about 10 m/s² against 1.6 m/s² on the Moon). Using $v = \sqrt{2gh}$, a larger g gives a larger v.
2023 Q8(a)	Paper sheet vs paper ball dropped from 100 m: complete the table.	**Mass:** equal. **Force exerted on the ground (weight):** equal. **Time taken to reach the ground:** sheet takes **more than** the ball. **Drag force:** sheet experiences **more than** the ball (larger surface area).
2024 Q3(a)	TWO factors that influence the law of universal gravitation.	(1) The **masses** of the two bodies ($F \propto m_1 m_2$). (2) The **distance between them** ($F \propto 1/r^2$).

2024 Q4(a)	Rain droplets are evenly spaced near the ground but irregular near the cloud. Explain.	Near the cloud the drops are still **accelerating** (weight greater than drag), so they spread out irregularly. After falling some distance they reach **terminal velocity** (drag = weight, net force zero) and fall at a constant speed, so they are evenly spaced.
2024 Q6(e)	Does velocity remain constant during free fall?	**No.** In free fall the body has a constant acceleration g (about 10 m/s²), so its velocity increases by about 10 m/s every second (until air resistance produces terminal velocity).
2024 Q7(e)	F = 500 N, r = 10×10^6 m, $M_{Earth} = 6 \times 10^{24}$ kg, G = 6.67×10^{-11}. Find the mass of the satellite.	$F = G M m / r^2 \Rightarrow m = F r^2 / (G M) = 500 \times (10^7)^2 / (6.67 \times 10^{-11} \times 6 \times 10^{24}) = 5 \times 10^{16} / 4.0 \times 10^{14} =$ **≈ 125 kg**.
2024 Q9(b)	Which graph shows a ball falling from a height to terminal velocity?	**Graph B**: the velocity rises quickly at first, the slope decreases as drag increases, and the graph becomes flat at a constant (terminal) velocity. Graph A would mean velocity keeps increasing (no drag); Graph C shows the velocity decreasing, which does not happen in a fall.
2025 Q4(a)	Fill in the speed table (0 s: 0; 1 s: 9.8; 2 s: ?; 3 s: ?; 4 s: 39.2) and conclude.	(i) **19.6 m/s** (ii) **29.4 m/s**. **Conclusion:** the speed of a falling body increases by 9.8 m/s every second, i.e. it has a constant acceleration g = 9.8 m/s² ($v = g t$).
2025 Q7(b)	TWO motion characteristics of a free-falling body.	(1) It moves with a **constant acceleration g (about 9.8 m/s²) downwards**. (2) The acceleration is **independent of mass**, so all bodies fall at the same rate (velocity increases uniformly with time).
2025 Q7(c)	Gravitational force between a 60 kg man and a 20 kg dog 0.50 m apart (G = 6.67×10^{-11}).	$F = G m_1 m_2 / r^2 = 6.67 \times 10^{-11} \times 60 \times 20 / (0.50)^2 = 6.67 \times 10^{-11} \times 1200 / 0.25 =$ **3.2×10^{-7} N** (very small).
2025 Q8(a)	A falling leaf: how do the net force and acceleration change before and after terminal velocity?	**Before:** weight (driving force) > drag, so the net force is downward and non-zero; it decreases as drag increases, so the acceleration is non-zero but decreasing. **After reaching terminal velocity:** drag = driving force, so the **net force = 0 and acceleration = 0**; the leaf falls at a constant velocity.

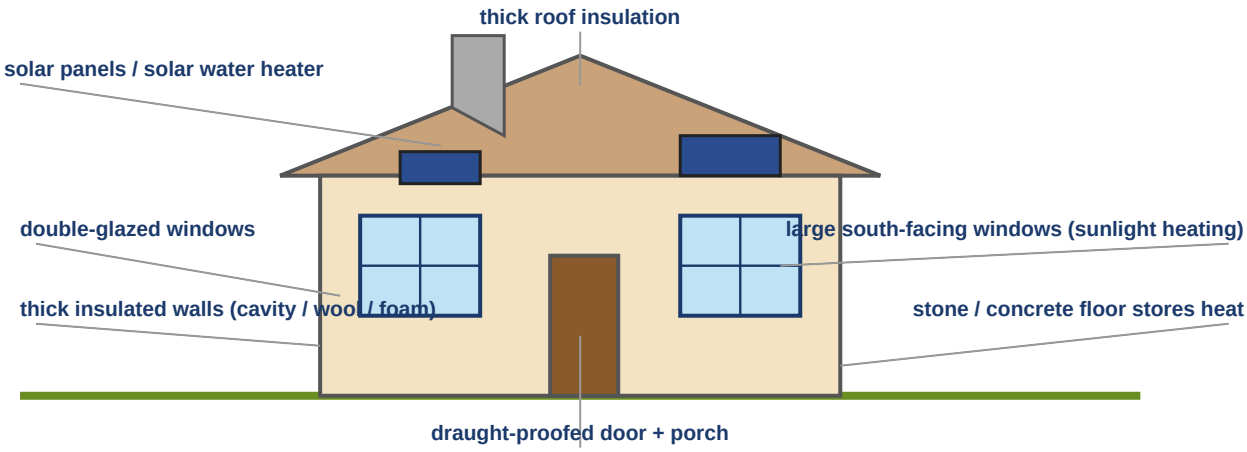
Chapter 3: Work, Energy, Power, Efficiency & Energy Resources

Calculations and "design / justify" questions in all three years

3.1 Must-know summary

Term	Meaning / rule
Work	Work = force × distance moved **in the direction of the force**. Unit: joule (J). **Zero work** is done when there is no displacement (pushing a wall) or when the force is perpendicular to the displacement (carrying a book while walking horizontally, a runner completing one lap of a circular track: displacement is zero). Work is **positive** if the force acts in the direction of motion and **negative** if it opposes it.
Energy	Capacity to do work (J). **Kinetic energy = ½ m v²**; **gravitational potential energy = m g h**. Law of conservation: energy cannot be created or destroyed, only changed from one form to another.
Power	Rate of doing work: P = W / t = F v. Unit: watt (W). The same work done in different times gives **different** powers.
Efficiency	Efficiency = (useful energy output / total energy input) × 100 %. It is always less than 100 % because some energy is wasted as heat and sound.
Energy efficiency	Doing the same job with **less energy**. Methods: efficient (5-star / LED) appliances, insulation, switching off, natural lighting, good design. Benefits: lower electricity bills, lower demand and imports, fewer emissions, resources conserved.
Voltage drop	The fall in voltage along wires because of their resistance (heat I^2R is wasted). Minimising it (thick, short wires; good joints; higher-voltage transmission) saves energy and keeps machines at full efficiency and safe.
Energy resources	Renewable (solar, wind, hydro) reduce dependence on fossil fuels and emissions. Bhutan depends on hydropower, which is low in winter, so solar adds energy security.

Formula	Use
W = F d P = W/t = F v KE = ½ m v² GPE = m g h	work, power, energies
Efficiency = (output / input) × 100 %	machines, appliances, see-saw, transformer



Also: LED lamps, 5-star appliances, thick curtains, heat-recovery stove

Fig 3.1 An energy-efficient house for a cold place (2023 Q8b)

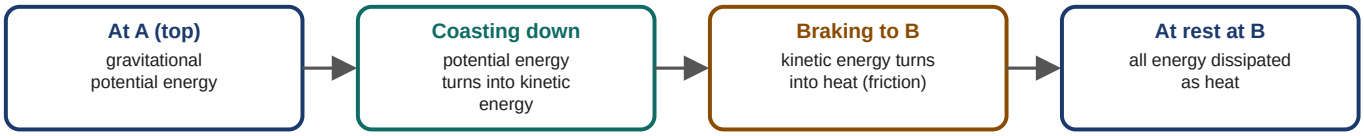


Fig 3.2 Energy changes of the cyclist (2025 MCQ-xvii)

3.2 Solved questions — Section A

Ref	Question	Ans	Why
2023 MCQ-iv	Which trimmer brand is the most efficient? (A 120 J in, 110 J out; B 110/100; C 112/105; D 105/100)	**D**	Efficiency: A 91.7 %, B 90.9 %, C 93.8 %, **D 95.2 %**.
2023 MCQ-xv	A 1 kg ball dropped from 10 m: which quantity is maximum just before it hits the ground?	**C**	Kinetic energy (PE has been converted into KE); weight and g are constant, acceleration is constant.
2023 MCQ-xix	Which is an example of zero work done?	**D**	An athlete completing a circular track returns to the start, so the displacement is zero (work = F × displacement = 0).
2024 MCQ-ii	Which situation gives zero work?	**C**	A boy walking horizontally with a book on his head: the force (weight/support) is perpendicular to the displacement.
2024 MCQ-xv	Work in ploughing 15 m with a force of 1000 N	**B**	W = F d = 1000 × 15 = **15 000 J**.
2024 MCQ-xxi	GPE of the 65 kg runner at Pangchenmo (4500 m)	**D**	GPE = m g h = 65 × 10 × 4500 = **2 925 000 J**.
2025 MCQ-iv	Most powerful athlete (200 m race): Nima 500 N, 18 s; Dawa 600 N, 20 s; Karma 700 N, 30 s; Mindu 800 N, 40 s	**B (Dawa)**	Same distance, so power ∝ weight / time: Nima 27.8, **Dawa 30**, Karma 23.3, Mindu 20.
2025 MCQ-xvii	Cyclist coasts from A and brakes to a stop at B: energy changes	**C**	gravitational potential → kinetic → heat (Fig 3.2).

Ref	Question	Answer and working
2023 T/F-ii	Energy efficiency leads to an increase in energy imports from other countries.	**FALSE** — it reduces energy use and therefore imports.
2023 Match-i	Five-star rated appliance	**e. energy efficiency**
2024 T/F-i	If the same work is done in different time intervals, the power is always the same.	**FALSE** — P = W/t: a shorter time gives more power.
2024 T/F-iii	Renewable sources such as solar or wind can decrease the reliance on fossil fuels.	**TRUE**
2025 Fill-iii	One of the best ways to reduce energy consumption for heating or cooling at home is…	**insulation** (of walls, roof and windows).
2025 T/F-i	For a machine, work done by the machine can never be greater than the work done on it.	**TRUE** — conservation of energy: output ≤ input (efficiency ≤ 100 %).
2025 Match-i	Product of force and velocity	**d. Power** (P = F v).

3.3 Solved questions — Section B

Ref	Question	Answer and working
2023 Q3(b)	A mother lifts a 15 kg baby to 2 m in 2 s. Find the power.	Work = m g h = 15 × 10 × 2 = 300 J. Power = W / t = 300 / 2 = **150 W** (147 W if g = 9.8).
2023 Q4(b)	Motorcycle A has a greater engine force than B but both reach the same velocity. Which has more power?	P = F v. With the same v, the larger force gives the larger power, so **motorcycle A has more power**.
2023 Q8(b)	Design an energy-efficient house for a cold place using methods of energy efficiency.	See Fig 3.1: **thick insulated walls and roof** (wool/foam/cavity); **double-glazed, small windows** on the cold side and **large south-facing windows** to gain sunlight; a draught-proofed door with a porch; **stone/concrete floor** that stores heat; **solar water heater/panels**; efficient heating (bukhari/heat recovery); LED lamps, 5-star appliances; thick curtains.
2023 Q8(e)	How does a high voltage drop in a circuit affect the efficiency of a machine?	A large voltage drop means much electrical energy is **wasted as heat in the wires** and the machine gets a lower voltage than needed, so its output is less and its **efficiency decreases**.
2023 Q9(c)	Bhutan should not depend on hydropower alone. Explain one alternative source with THREE minimal environmental impacts.	**Solar energy.** (1) No greenhouse gases or air pollution when generating electricity. (2) Rooftop panels need no dam or new land, so rivers, forests and wildlife are not disturbed. (3) No fuel, no waste and no noise during operation; it is renewable and complements hydropower in winter when river flow is low.
2024 Q2(a)	TWO benefits of energy-efficient appliances.	(1) They **use less electricity**, so bills are lower. (2) Lower energy demand means **fewer emissions and less use of natural resources** (and a longer life for the appliance).
2024 Q4(b)	Significance of minimising voltage drop in an electric design.	It **reduces energy loss as heat** in the wires, keeps the supply voltage at the appliances correct so they work properly, avoids overheating and improves efficiency.
2024 Q4(c)	Design an energy-efficient classroom.	**Efficient appliances:** LED lamps, 5-star fans and computers, occupancy/motion sensors and switches. **Insulation:** insulated roof and walls, double-glazed windows, draught-proofed doors. **Technology:** solar panels / solar water heater, smart meters, timers. **Design:** large windows for daylight and cross-ventilation, light-coloured walls, trees for shade, rainwater harvesting. **Result:** less energy and waste, resources conserved, minimal environmental impact.
2024 Q5(b)	A 300 W electric wheelchair (50 kg) with a 50 kg occupant climbs a 10 m high ramp at steady velocity. Time?	Work = m g h = (50 + 50) × 10 × 10 = 10 000 J. Time = W / P = 10 000 / 300 = **33.3 s**.
2024 Q7(c)	Tug of war: Team A pulls 200 N, Team B 1000 N. Which team did positive work?	**Team B**: its force acts in the direction of the displacement of the rope (the rope moves towards B). Team A's force opposes the motion, so it does negative work.
2024 Q9(d)	A 1200 kg car at 15 m/s brakes and its speed is halved. Find the KE.	New speed = 7.5 m/s. KE = ½ m v^2 = 0.5 × 1200 × 7.5^2 = 600 × 56.25 = **33 750 J**.
2025 Q2(b)	A 250 g ball is thrown up at 30 m/s; find its KE at 2 s using the graph.	From the graph, v at 2 s = **10 m/s**. KE = ½ m v^2 = 0.5 × 0.25 × 10^2 = **12.5 J**.
2025 Q5(a)	A man climbs a slope; another walks the same distance on level ground. Who expends more energy?	The **man climbing the slope**: besides overcoming friction he does work against gravity, raising his weight through a height (gaining m g h of potential energy).

2025 Q7(f)	Why is Bhutan investing in solar power (250 kW and 500 kW plants)?	Hydropower is **low in winter (dry season)**, forcing imports; solar is strong then, so it gives **energy security** and reduces imports. Solar is clean and renewable (supports carbon neutrality), quick to build, suits rooftops and rural areas, and diversifies energy sources.
2025 Q9(b)	See-saw: A = 750 N, B = 450 N; B is lifted 0.5 m when A sits down. Efficiency?	Equal arms, so A moves down 0.5 m. Work output (lifting B) = 450 × 0.5 = 225 J. Work input (A's weight) = 750 × 0.5 = 375 J. Efficiency = 225 / 375 × 100 = **60 %**.

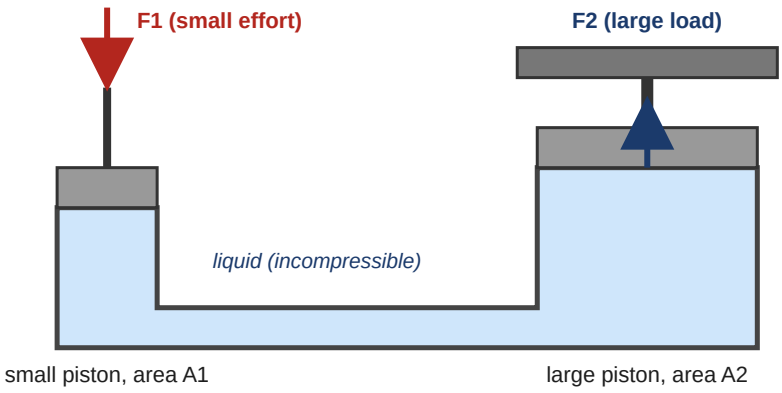
Chapter 4: Pressure, Pascal's Law & Hydraulic Machines

Every year: at least one hydraulic calculation and several pressure MCQs

4.1 Must-know summary

Term	Meaning / rule
Pressure	Force acting **perpendicularly per unit area**: **P = F / A**. Unit: pascal (Pa) = N/m².
Pressure due to solids	Depends on (1) the **force (weight)** and (2) the **area of contact**. P ∝ F and P ∝ 1/A. Same force on a smaller area → larger pressure (sharp knife, nail, rocket nozzle). Larger area → smaller pressure (wide bag strap, camel's feet, wide foundations, tractor tyres).
Pressure in liquids	**P = ρ g h**: increases with **depth** and **density**; acts in all directions; same at the same level, independent of the shape of the container. Sea water is denser than river water, so the pressure at the same depth is **greater**.
Atmospheric / water pressure change	As a bubble rises, the water pressure on it falls, so it **expands**.
Pascal's law	Pressure applied to an **enclosed fluid** is transmitted **equally and undiminished in all directions** through the fluid and to the walls of the container.
Hydraulic machine	Small effort force F_1 on a small piston (A_1) gives pressure $P = F_1/A_1$, transmitted to a large piston (A_2): $\mathbf{F_2 = F_1 \times A_2/A_1}$. It is a **force multiplier** (jack, brakes, lifts, excavators). Work in = work out, so the small piston moves a longer distance.
Gas in place of liquid	Pascal's law holds for any fluid, including gases, but a gas is **compressible**: part of the effort is used in compressing it, so it is unsuitable for hydraulic machines. Liquids (oil) are used.

Formula	Use
P = F / A	pressure of solids; bricks, shoes, knife, rocket nozzle
P = ρ g h	hydrostatic pressure (submarine, diver)
$F_1/A_1 = F_2/A_2$	hydraulic machines



Pascal's law: P1 = P2 => F1 / A1 = F2 / A2 => F2 = F1 x (A2 / A1)

A2 > A1 => F2 > F1 : the machine is a FORCE MULTIPLIER (work in = work out: F1 d1 = F2 d2)

Fig 4.1 A hydraulic machine: Pascal's law and the force multiplier

4.2 Solved questions — Section A

Ref	Question	Ans	Why
2023 MCQ-v	Sangay's shoe is smaller than Dema's but both exert the same pressure on the ground. Choose the correct statement.	**B**	P = F/A. Same P with the smaller area means a smaller force: **Dema is heavier** than Sangay.
2023 MCQ-vi	A hydraulic machine uses a gas instead of a liquid (F_1 = 10 N, A_1 = 1 m², A_2 = 5 m²). Which is correct?	**C**	Pascal's law still applies: the pressure at pistons X and Y is equal. Because a gas is compressible, the output force cannot be relied upon to be the ideal 50 N, so statements about F_2 (A, B) are not correct.
2024 MCQ-i	Main advantage of a hydraulic system	**B**	Multiplying force.
2024 MCQ-xi	Which school bag is most comfortable?	**D**	A wide strap of soft canvas: a larger area gives a smaller pressure on the shoulder.
2024 MCQ-xii	Force at area 1 compared with area 2 in the hydraulic lift	**A**	The pressure is the same; the small piston (area 1) needs a smaller force than the large piston (area 2).
2024 MCQ-xxii	Rocket thrust 10 000 N, nozzle area 2 m²: pressure	**B**	P = F/A = 10 000 / 2 = **5000 Pa**.
2025 MCQ-v	A brick is turned onto its smallest face. Effect on force and pressure?	**C**	The force (weight) is **unchanged**, the area is smaller, so the pressure **increases**.
2025 MCQ-vi	Equation for hydrostatic pressure	**C**	P = ρ g h.
2025 MCQ-xviii	Why do bubbles grow as they rise?	**B**	The water pressure on the bubble decreases as the depth decreases, so the gas inside expands.
2025 MCQ-xix	Working principle of a hydraulic machine	**D**	It applies the same amount of pressure everywhere (to the load), which gives a larger force on the larger piston.

Ref	Question	Answer and working
2023 Fill-ii	The pressure exerted at equal depth in sea water is _____ than in a river.	**greater** (sea water is denser).
2023 Match-ii	Explains the magnitude of pressure in a confined liquid	**c. Pascal's law**

4.3 Solved questions — Section B

Ref	Question	Answer and working
2023 Q4(c)	8 bricks (0.2 m × 0.1 m × 0.1 m, 60 N each) lie in a single layer. Minimum pressure on the ground?	Minimum pressure means the **largest area**: each brick lies on its 0.2 × 0.1 = 0.02 m² face. Total area = 8 × 0.02 = 0.16 m²; total weight = 8 × 60 = 480 N. P = F/A = 480 / 0.16 = **3000 Pa**.
2023 Q5(c)	Hydraulic lift: piston area 0.6 m² with force 40 N lifts a 1000 N load. Area of the other piston?	$F_1/A_1 = F_2/A_2 \Rightarrow A_2 = F_2 A_1 / F_1$ = 1000 × 0.6 / 40 = **15 m²**.
2023 Q6(b)	TWO factors affecting the pressure due to solids.	(1) The **force (weight) acting on the surface**. (2) The **area of contact** (P = F/A).

2023 Q7(c)	Four pipes join the same tank at the same level. Is the pressure equal in all four houses? Why?	**Yes.** Pressure in a liquid depends only on depth, density and g ($P = \rho g h$), not on the length or direction of the pipes. All four pipes are at the same depth, so the pressure is equal.
2023 Q8(c)	Explain with a mathematical expression why hydraulic machines act as force multipliers.	By Pascal's law $P_1 = P_2$, so $F_1/A_1 = F_2/A_2 \Rightarrow$ **$F_2 = F_1 (A_2/A_1)$**. If $A_2 > A_1$, then $F_2 > F_1$: the force is multiplied by the ratio of the areas.
2024 Q2(b)	How does Pascal's principle apply to hydraulic jacks?	A small force on the small piston creates a pressure $P = F_1/A_1$ in the oil. The pressure is **transmitted equally throughout the liquid** and acts on the large piston, giving a large force $F_2 = P A_2$ that lifts the car.
2024 Q3(c)	Does Pascal's principle hold for a gas? Reason?	**Yes**: a gas is a fluid, and the pressure applied to an enclosed gas is also transmitted equally in all directions. (But a gas is compressible, so it is not used in hydraulic machines.)
2024 Q6(a)	A pump lifts a 1000 N crate; the small piston (50 cm²) applies 500 N. Area of the larger piston?	$A_2 = F_2 A_1 / F_1 = 1000 \times 50 / 500 =$ **100 cm²**.
2024 Q6(b)(i)	Factors contributing to the Titan submersible implosion at 3500 m.	(1) The **enormous water pressure** at 3500 m: $P = \rho g h \approx 1030 \times 10 \times 3500 \approx 3.6 \times 10^7$ Pa (about 350 times atmospheric pressure). (2) A **weakness of the hull** (composite material, fatigue from repeated dives, a design/testing flaw) that could not withstand this pressure.
2024 Q6(b)(ii)	Why an implosion rather than an explosion?	The **pressure outside (water) was far greater than the pressure inside** (about 1 atmosphere), so the hull collapsed inwards. An explosion happens when the pressure inside is much greater than outside.
2025 Q3(b)	$F_1 = 306$ N on a 2 m² piston. Pressure on the large piston?	$P_1 = F_1/A_1 = 306 / 2 = 153$ Pa. By Pascal's law the pressure is the same everywhere, so $P_2 =$ **153 Pa** (N/m²).
2025 Q5(c)	Explain the working of a hydraulic jack.	A small force on the small piston produces pressure $P = F_1/A_1$ in the liquid. By Pascal's law this pressure is transmitted undiminished to the large piston of area A_2, which therefore exerts a larger force $F_2 = P A_2 = F_1 A_2/A_1$ that lifts the car (the small piston moves farther, so work in = work out).
2025 Q6(c)	Relationship between surface area and pressure (sharp knife).	For the same force, pressure is **inversely proportional to the surface area** ($P = F/A$): the sharp edge has a small area, so a large pressure cuts easily.
2025 Q8(b)	A 300 kg wooden block, 5 m × 0.5 m, lies flat. Pressure on the ground?	Weight = 300 × 10 = 3000 N; area = 5 × 0.5 = 2.5 m². $P = 3000 / 2.5 =$ **1200 Pa** (1176 Pa if g = 9.8).
2025 Q9(c)	Explain Pascal's law.	**Pressure applied to an enclosed fluid is transmitted equally and undiminished in all directions** throughout the fluid and to the walls of its container.

Chapter 5: Heat, Specific Heat Capacity & Calorimetry

A specific-heat calculation appears every year

5.1 Must-know summary

Term	Meaning / rule
Heat vs temperature	Heat is energy transferred (J); temperature shows how hot a body is and **decides the direction of heat flow**: from the hotter to the colder body. Two bodies can have the same temperature but different heat contents (a bucket and a cup of water).
Thermal equilibrium	When two bodies in contact reach the **same temperature** there is no net heat flow.
Specific heat capacity (c)	The heat needed to raise the temperature of **1 kg** (or 1 g) of a substance by **1 °C**. Water: **4200 J/kg°C** (4.2 J/g°C, 1 cal/g°C) — very high.
Heat capacity	Heat needed to raise the temperature of a given body by 1 °C = m c. Thicker (more massive) pan bases have a larger heat capacity.
Latent heat	Heat absorbed or released during a **change of state at constant temperature**: latent heat of fusion (ice → water, 360–334 J/g), of vaporisation (water → steam). Boiling water stays at 100 °C until the last drop evaporates.
Principle of calorimetry	With no heat lost to the surroundings: **heat lost by the hot body = heat gained by the cold body** (law of conservation of energy). A calorimeter is a well-insulated vessel (Fig 5.2).
Applications of high s.h.c. of water	Coolant in car radiators, hot-water bags, fomentation, coastal climate, watering plants before frost (water releases heat slowly and latent heat when freezing).

Formula	Use
$Q = m\,c\,\Delta T$	heat gained or lost; $c = Q / (m\,\Delta T)$; same Q ⇒ smallest ΔT for the largest c
$Q = m\,L$	change of state (latent heat)
$m_1c_1(T_1 - T) = m_2c_2(T - T_2)$	mixing / calorimetry problems

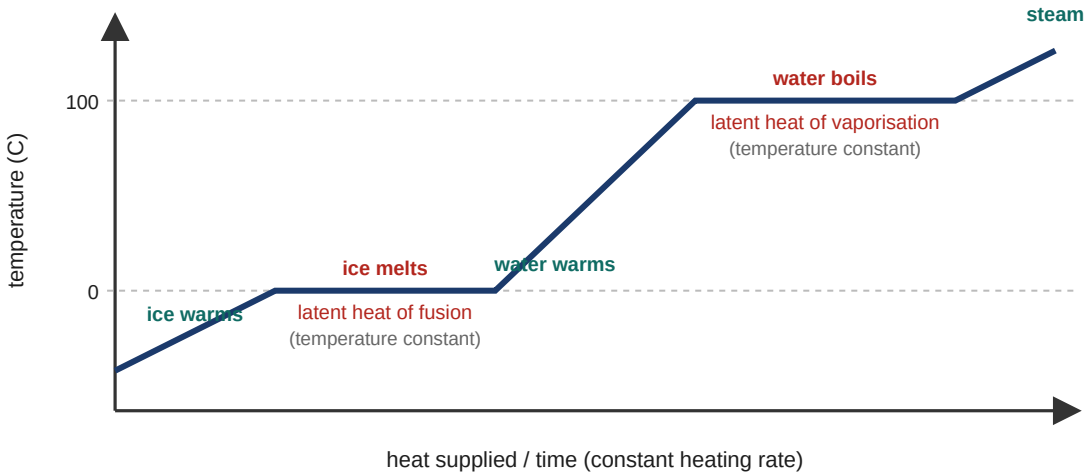


Fig 5.1 Heating curve of ice: the flat parts are latent heat (change of state)

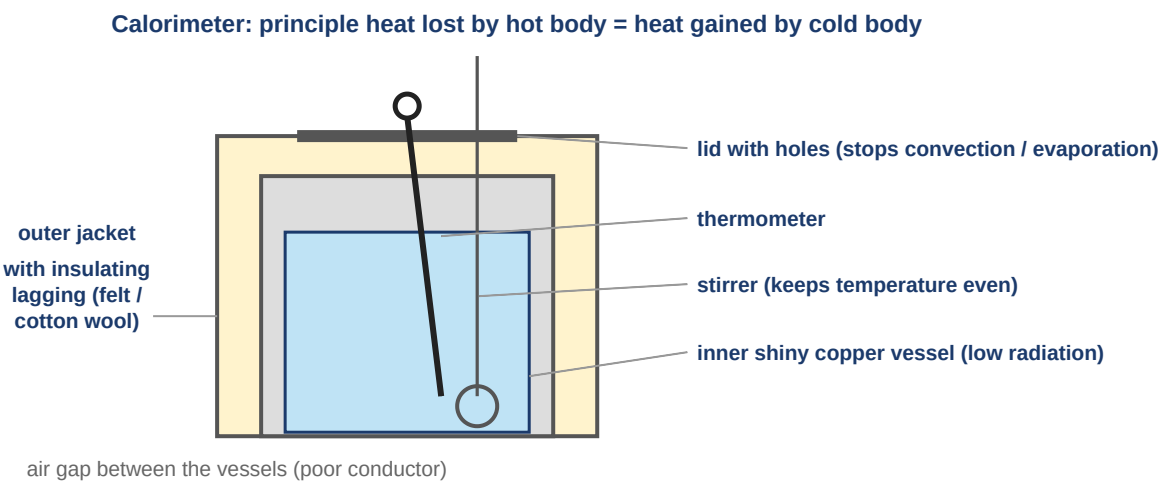


Fig 5.2 A calorimeter and how to make it lose (almost) no heat

5.2 Solved questions — Section A

Ref	Question	Ans	Why
2023 MCQ-vii	A cold and a hot body are kept in contact. What happens?	**C**	Heat flows from the hot body to the cold body.
2023 MCQ-xvi	2 kg of water heated from 20 to 80 °C: heat gained?	**C**	Q = m c ΔT = 2 × 4200 × 60 = **504 000 J**.
2023 MCQ-xxv	Equal heat to 50 g each of copper (0.38), silver (0.24), iron (0.46), lead (0.13 J/g°C): which changes temperature least?	**B (iron)**	ΔT = Q/(mc): same Q and m, so the largest c (iron, 0.46) gives the smallest rise.
2024 MCQ-v	No heat flows between two bodies A and B. This suggests…	**A**	both bodies have the same temperature (thermal equilibrium).
2025 MCQ-vii	25 kg copper bar heated from 25 to 55 °C (c = 385 J/kg°C)	**C**	Q = 25 × 385 × 30 = **288 750 J**.
2025 MCQ-xx	Graph for ice at −10 °C heated steadily to water at +10 °C	**B (X)**	Temperature rises to 0 °C, stays **flat** while the ice melts (latent heat), then rises again.

Ref	Question	Answer and working
2023 Fill-iii	When two bodies in contact attain the same temperature, they are in _____ equilibrium.	**thermal**
2023 Match-iii	The temperature of boiling water remains 100 °C till the last drop evaporates	**a. latent heat of vaporisation**
2024 Fill-iii	There is no change in _____ when a substance changes state (solid → liquid or liquid → gas).	**temperature**
2024 T/F-v	If equal heat raises liquid A's temperature more than liquid B's, A is more effective as a heat reservoir.	**FALSE** — A has the *lower* specific heat capacity; B (higher c) stores more heat and is the better reservoir.
2024 Match-iii	High specific heat capacity of water	**a. fomentation** (hot-water bags)
2024 Match-v	Base of cooking pans are thicker	**g. increased heat capacity**

2025 Fill-iv	The direction of heat transfer between two bodies is determined by…	**temperature** (heat flows from higher to lower temperature).
2025 T/F-ii	Two bodies can have the same temperature even if their heat contents are different.	**TRUE**
2025 Match-ii	Snow on the mountain does not melt at once	**e. specific latent heat of fusion of water**

5.3 Solved questions — Section B

Ref	Question	Answer and working
2023 Q3(c)	The principle of calorimetry and the law of conservation of energy are similar. Agree? Justify.	**Agree.** Calorimetry states heat lost by the hot body equals heat gained by the cold body. This is the conservation of energy applied to heat: the energy is not created or destroyed, it is only transferred from one body to the other.
2023 Q5(d)(i)	Define specific heat capacity.	The **amount of heat required to raise the temperature of 1 kg (or 1 g) of a substance by 1 °C** (unit J/kg°C).
2023 Q5(d)(ii)	Why do ice cubes cool a juice better than water at 0 °C?	Ice must absorb the **latent heat of fusion** to melt, in addition to the heat it gains afterwards. Cold water at 0 °C has no latent heat to absorb, so ice removes much more heat from the juice per gram.
2023 Q6(c)(i)	Heat needed to raise 10 kg of water by 1 °C.	$Q = m\,c\,\Delta T = 10 \times 4200 \times 1 =$ **42 000 J** (= 10 000 cal).
2023 Q6(c)(ii)	5 kg of water at 80 °C is mixed with 10 kg of cold water; the final temperature is 40 °C. Initial temperature of the cold water?	Heat lost by hot water = $5 \times 4200 \times (80 - 40) = 840\,000$ J. Heat gained by cold water = $10 \times 4200 \times (40 - T)$. Equate: $42\,000\,(40 - T) = 840\,000 \Rightarrow 40 - T = 20 \Rightarrow T =$ **20 °C**.
2023 Q8(d)	1 g of ice at 0 °C absorbs 360 J to melt. Heat to bring 1000 g of ice to boiling point (c = 4.2 J/g°C)?	Melting: $1000 \times 360 = 360\,000$ J. Heating water 0 → 100 °C: $1000 \times 4.2 \times 100 = 420\,000$ J. Total = **780 000 J (780 kJ)**.
2024 Q3(d)	How to modify the calorimeter for zero heat loss?	See Fig 5.2: put the inner (shiny, polished) vessel inside an **outer jacket with insulating lagging** (felt, cotton wool, polystyrene) leaving an air gap; **cover it with an insulating lid** with holes for the thermometer and stirrer; the vacuum flask (silvered double wall with vacuum) is the ideal.
2024 Q4(e)	A 0.05 kg metal piece needs 2.4×10^3 J to go from 27 to 327 °C. Specific heat capacity?	$\Delta T = 300$ °C. $c = Q / (m\,\Delta T) = 2400 / (0.05 \times 300) = 2400 / 15 =$ **160 J/kg°C**.
2024 Q5(a)	Heat lost by A = 500 J; heat gained by B = 450 J. Was the principle of calorimetry confirmed?	**No**: the principle requires heat lost = heat gained. The 50 J difference was lost to the surroundings/calorimeter (the experiment was not perfectly insulated).
2024 Q6(d)	Graph of temperature rise of liquids A, B, C with time: which is the best radiator coolant?	**Liquid C**: for the same heating time its temperature rises least, so it has the **highest specific heat capacity** and can absorb the most heat from the engine for a small rise in temperature.
2024 Q8(d)	Which cools a soft drink faster: ice cubes at 0 °C or iced water at 0 °C?	**Ice cubes**: as they melt they absorb the latent heat of fusion from the drink in addition to ordinary heat, so they take more heat per gram than water at 0 °C.
2025 Q2(c)	A farmer watered his vegetables before the frost. Why?	Water has a **high specific heat capacity**, so the wet soil and water cool slowly, and when water freezes it **releases latent heat**; this keeps the plants' temperature near 0 °C and protects them from frost damage.

2025 Q4(b)	8400 J heats 0.08 kg of metal (c = 1050 J/kg°C) from 30 °C. Final temperature?	$Q = m\,c\,(T - 30) \Rightarrow 8400 = 0.08 \times 1050 \times (T - 30) = 84\,(T - 30) \Rightarrow T - 30 = 100 \Rightarrow T =$ **130 °C**.
2025 Q4(e)	Why does ice relieve the pain of a burn?	Heat flows from the hot burnt skin to the colder ice, so the **skin temperature falls**; the melting ice absorbs latent heat and the cold also numbs the nerve endings, reducing pain and swelling.
2025 Q8(c)	Why does the temperature drop after a sudden downpour in hot summer?	The cooler rain water absorbs heat from the hot ground and air, and its **evaporation takes latent heat from the surroundings**; the clouds also block sunlight. Together they lower the temperature.
2025 Q9(e)	How does a calorimeter apply the law of conservation of energy?	In an insulated calorimeter no heat escapes, so **heat lost by the hot substance = heat gained by the cold substance and the vessel**: $m_1c_1(T_1 - T) = m_2c_2(T - T_2)$. Energy is only transferred, never lost, which is used to find the specific heat capacity of an unknown substance.

Chapter 6: Current Electricity — Ohm's Law, Resistance & Heating Effect

V–I graphs and a current / power calculation appear every year

6.1 Must-know summary

Term	Meaning / rule
Electric current (I)	The rate of flow of **electric charge** in a circuit: I = Q / t. Unit: ampere (A). Conventional current flows from the positive to the negative terminal; electrons flow from the negative (higher electron concentration) towards the positive terminal.
Potential difference (V)	Energy given per unit charge; unit: volt (V).
Resistance (R)	Opposition to current: R = V / I; unit: ohm (Ω). R depends on **length** (R ∝ L), **cross-sectional area** (R ∝ 1/A: a thicker wire has less resistance), **material** and **temperature**.
Ohm's law	At constant temperature (and other physical conditions) the current through a conductor is directly proportional to the potential difference: V ∝ I, i.e. **V / I = R is constant**. The V–I graph is a **straight line through the origin**; the slope of V–I = R. Triangle: V = I × R.
Ohmic and non-ohmic	Ohmic: copper, nichrome (constant V/I). Non-ohmic: filament lamp, diode, thermistor, tungsten at high current (V/I changes) — "variable resistance".
Heating effect	H = I² R t: depends on **current**, **resistance** and **time**. A filament lamp has a thin, high-resistance tungsten filament, so it becomes white/red hot.
Electrical power	P = V I = I² R = V² / R. Current drawn: I = P / V.

Formula	Use
V = I R I = Q / t P = V I	circuit calculations
H = I² R t	heat produced

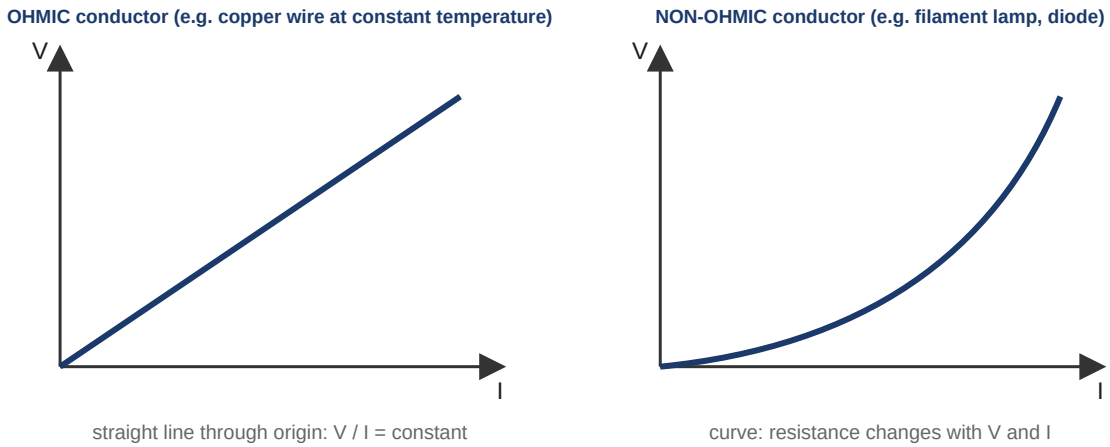


Fig 6.1 V-I graphs of an ohmic and a non-ohmic conductor

6.2 Solved questions — Section A

Ref	Question	Ans	Why
2023 MCQ-xvii	V–I graph for conductors L, M, N (L steepest, M curved, N gentle). Which statement is correct?	**D**	The slope V/I = R: L has the steepest line, so **L has the maximum resistance**; M is non-ohmic (curved).
2024 MCQ-xix	Which are the ohmic conductors in the table (V and I readings)?	**C**	Copper: V/I = 20 Ω every time; nichrome: V/I = 10 Ω every time → **copper and nichrome**. Aluminium and tungsten have changing V/I.
2025 MCQ-viii	V–I graph (current on y-axis) for R_1–R_4: relation between the resistances?	**B**	With I on the y-axis, a **steeper** line means less resistance. R_3 is steepest and R_2 flattest: **$R_2 > R_4 > R_1 > R_3$**.
2025 MCQ-xxi	The triangle V over I and R represents which law?	**A**	Ohm's law: V = I R.

Ref	Question	Answer and working
2023 Fill-iv	Electric current is the rate of flow of _____ in a circuit.	**charge** (electric charge).
2023 T/F-iii	The resistance of a conductor will double if its thickness is doubled.	**FALSE** — resistance is inversely proportional to the cross-sectional area; a thicker wire has *less* resistance.
2024 Fill-v	The resistance of a wire decreases when its cross-sectional area _____.	**increases**
2024 Match-i	Non-ohmic conductor	**d. variable resistance**
2025 T/F-iii	Electric current flows from the region of higher concentration of electrons to the region of lower concentration across a conductor.	**TRUE** (electron flow: from the negative terminal towards the positive terminal; the conventional current is drawn in the opposite direction).

6.3 Solved questions — Section B

Ref	Question	Answer and working
2023 Q3(d)	TWO factors on which the heat produced by the heating effect of current depends.	H = I² R t, so it depends on (1) the **current (I)**, (2) the **resistance (R)** (and the time t).
2023 Q7(d)	220 V supply; 100 W incandescent bulbs and 9 W LED bulbs. Current drawn by each?	I = P / V. Incandescent: 100 / 220 = **0.45 A**. LED: 9 / 220 = **0.04 A**. (The LED draws about 11 times less current.)
2024 Q2(d)	Graph for wires A and B: which obeys Ohm's law? What must stay constant?	**Wire B** (straight line through the origin: V ∝ I). Wire A is curved (non-ohmic). To verify Ohm's law the **temperature** (and the length, cross-section and material) of the conductor must remain constant.
2024 Q2(e)	An 8 Ω resistor with a 20 V battery: ammeter reading?	I = V / R = 20 / 8 = **2.5 A**.
2025 Q3(c)	Plot R against I for the data (R = 0, 10, 20, 30, 40 Ω; I = 100, 80, 60, 40, 20 A) and state the relationship.	See Fig 6.2: the points lie on a **straight line with a negative slope** (I = 100 − 2R). **As the resistance increases, the current decreases** (they are inversely related; the slope is −2 A/Ω).
2025 Q6(b)	Hair dryer resistance 55 Ω at 220 V: wattage?	P = V² / R = 220² / 55 = 48 400 / 55 = **880 W** (I = 4 A).

2025 Q9(d)	Why does the filament of a lamp become red hot?	The filament is a **thin tungsten wire of high resistance**; when current flows, electrical energy is converted to heat ($H = I^2Rt$), raising its temperature until it glows.

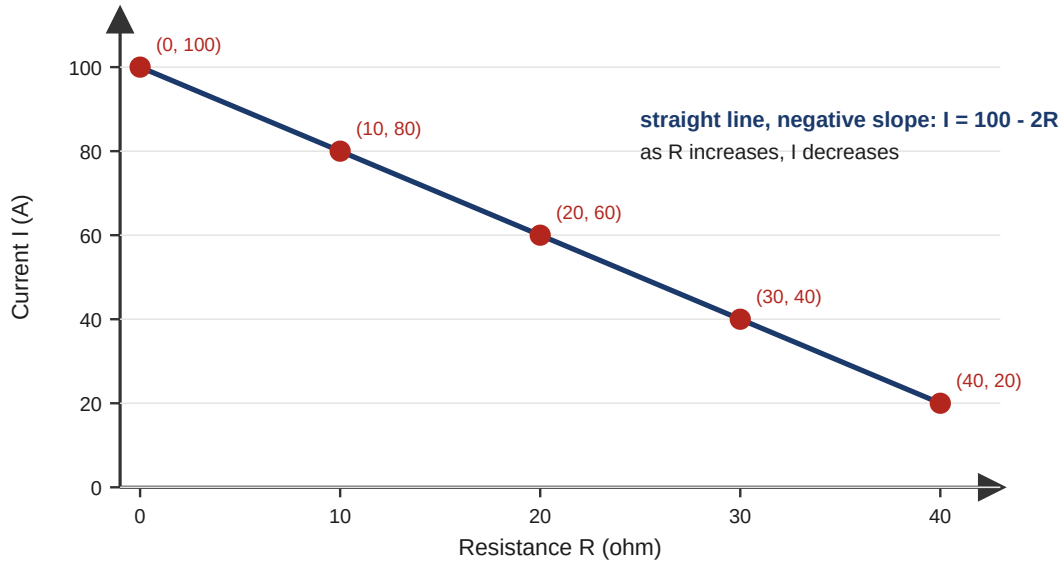


Fig 6.2 Plot of resistance against current for 2025 Q3(c)

Chapter 7: Electromagnetism, Generators & Transformers

A transformer calculation and a Fleming-rule question appear every year

7.1 Must-know summary

Term	Meaning / rule
Motor effect	A current-carrying conductor placed in a magnetic field experiences a **force**. Direction by **Fleming's left-hand rule**: **Thumb = Force (motion), First finger = Field (N→S), Second finger = Current**. Reversing the current (or the poles) reverses the direction of the force/rotation; reversing both leaves it unchanged. Used in the electric motor.
Electromagnetic induction	An emf (and current) is induced in a coil whenever the **magnetic flux through it changes** (magnet moved in or out of a solenoid, a conductor cutting field lines). Faraday's law. Direction by **Fleming's right-hand rule**: **Thumb = Motion, First finger = Field, Second finger = Induced current**.
Generator	Rotating a coil in a magnetic field induces an emf. **AC generator: slip rings** and carbon brushes → alternating current. **DC generator:** a **split-ring commutator** → direct current. The emf is greatest when the coil is parallel to the field (cutting lines fastest).
Transformer	Changes AC voltage using two coils on an iron core. $\mathbf{V_s/V_p = N_s/N_p}$. Step-up: $N_s > N_p$; step-down: $N_s < N_p$ (mobile charger). Ideal (100 %): $V_pI_p = V_sI_s$, so voltage up means current down. Works only with AC.
Why not 100 % efficient	Heat in the coils (I^2R, copper loss), **eddy currents** in the core, hysteresis loss, **flux leakage**, and sound/vibration.
Step-up for transmission	The same power P = V I is sent at a very high voltage, so the current is small and the heat loss I^2R in the long lines is small. A step-down transformer then lowers the voltage for homes.

Formula	Use
$V_s/V_p = N_s/N_p$	voltage / turns
$V_pI_p = V_sI_s$	currents of an ideal transformer

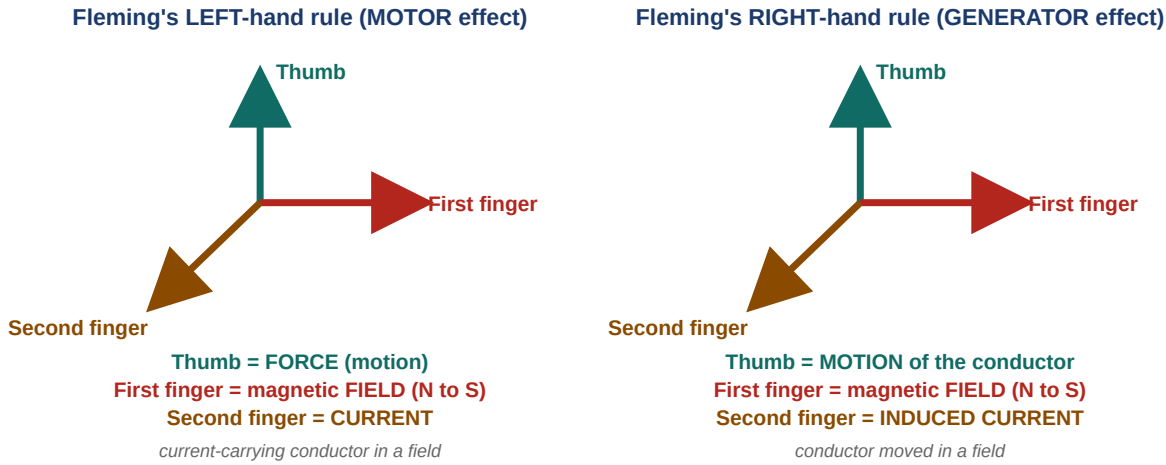


Fig 7.1 Fleming's left-hand rule (motor) and right-hand rule (generator)

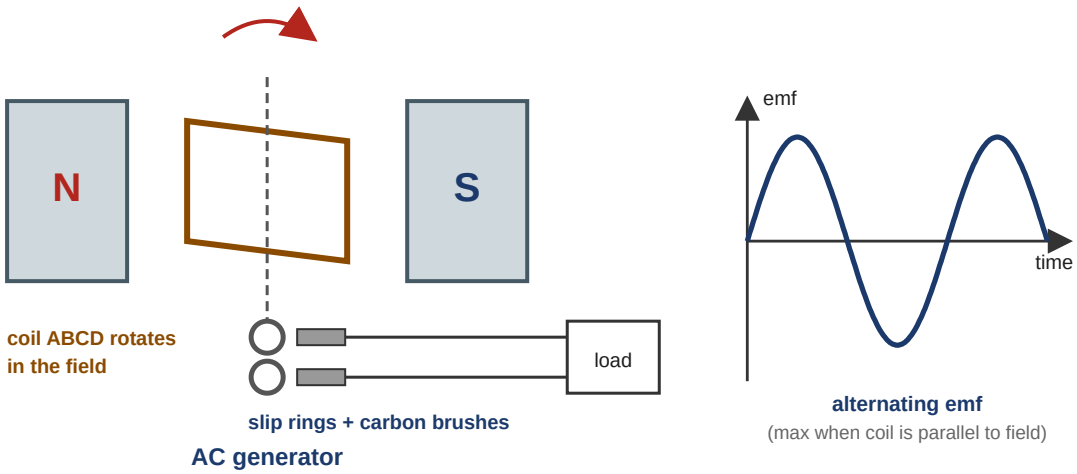
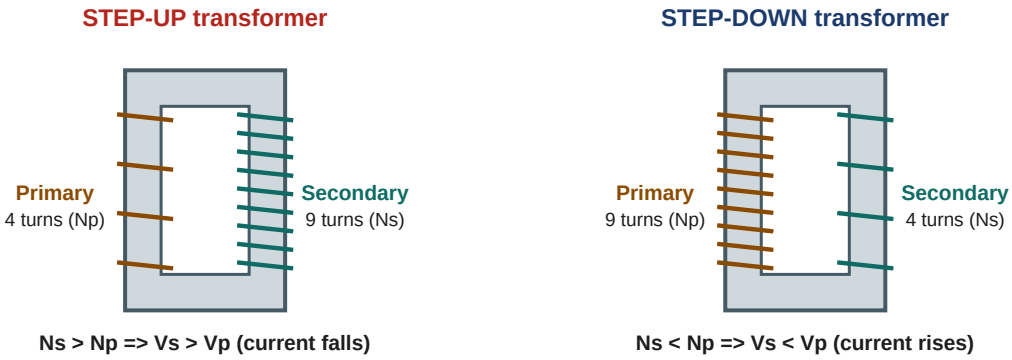


Fig 7.2 An AC generator and its alternating emf



Vs / Vp = Ns / Np and, for an ideal (100% efficient) transformer: Vp Ip = Vs Is

Works only with AC: a changing current in the primary changes the magnetic flux in the iron core and induces an emf in the secondary.

Fig 7.3 Step-up and step-down transformers

7.2 Solved questions — Section A

Ref	Question	Ans	Why
2023 MCQ-viii	Mobile charger transformer: which statement describes it? (5 V needed from the mains)	D	It is a **step-down** transformer: the secondary coil has fewer turns than the primary.
2023 MCQ-xviii	A transformer lowers 220 V to 50 V; input current 0.5 A. Output current?	C	$V_pI_p = V_sI_s \Rightarrow I_s = 220 \times 0.5 / 50 =$ **2.2 A**.
2024 MCQ-xx	A coil rotates in a field when current flows. What makes it rotate the opposite way?	C	Reversing the direction of the current (reversing both current and poles would give the same direction again).
2024 MCQ-xxv	Device based on Faraday's law of electromagnetic induction	D	The electric generator (the motor works on the motor effect).
2025 MCQ-ix	A bar magnet moved in and out of a solenoid deflects a galvanometer. Concept shown?	D	The induced current is due to the **change in magnetic flux**.
2025 MCQ-xxii	Fundamental difference between an AC and a DC generator	D	The AC generator has **slip rings**; the DC generator has a **split-ring commutator**.

Ref	Question	Answer and working
2023 Match-iv	Principle of electromagnetic induction	**No option fits.** Column B does not contain Faraday's law (the options are latent heats, infrared, Pascal's law, energy efficiency, radio waves, Ohm's law), so this item is faulty. The correct match would be Faraday's law of electromagnetic induction; Ohm's law (g) is the only electrical option but it is not related to induction.
2024 Match-iv	Device that transfers electrical energy between two or more circuits	**b. transformer**

7.3 Solved questions — Section B

Ref	Question	Answer and working
2023 Q2(b)(i)	Generator diagram (N pole left, S pole right; coil ABCD rotating as shown): direction of induced current using Fleming's right-hand rule.	Forefinger along the field (N → S), thumb along the motion of the sides (the side near S moves towards the viewer), centre finger gives the current: **D → C → B → A** (up the right-hand side, across the top, down the left side). If the coil rotated the other way, the direction would reverse. (Use the arrow in your paper to decide.)
2023 Q2(b)(ii)	Name parts X and Y.	**X: slip rings** (two rings, each joined to one end of the coil); **Y: carbon brushes** (spring-loaded contacts pressing on the rings that connect to the external circuit).
2023 Q4(d)(i)	The picture shows Fleming's right-hand rule. Identify the error.	The labels are wrong: the **thumb should show the motion of the conductor and the centre (second) finger the induced current**, but the diagram labels the thumb "induced current" and the centre finger "motion of the conductor".
2023 Q4(d)(ii)	Name the rule.	**Fleming's right-hand rule** (generator rule).
2023 Q5(e)	Identify the transformers in Figures A and B.	**A:** the secondary coil has more turns than the primary, so it is a **step-up transformer**. **B:** the primary has more turns than the secondary: a **step-down transformer**.
2023 Q9(d)	Table: transformer A N_s:N_p = 10:1, V_p = 16 V, find Q = V_s; transformer B N_s:N_p = 12:3, V_s = 60 V, find R = V_p.	A: $V_s = V_p \times (N_s/N_p) = 16 \times 10 =$ **Q = 160 V**. B: $N_s/N_p = 12/3 = 4$, so $V_p = 60 / 4 =$ **R = 15 V**.
2024 Q6(c)	Transformer A or B for a higher voltage than the BPC supply?	**Transformer B**: its secondary coil has more turns than the primary, so it is a step-up transformer ($V_s > V_p$). A has fewer turns in the secondary (step-down).
2024 Q7(a)	AC generator: coil ABCD (suspended between concave pole pieces, parallel to the field, rotating as shown). Direction of current and the rule?	By **Fleming's right-hand rule**: with the field from N (left) to S (right) and the side DC moving towards the viewer (clockwise rotation seen from above), the current in DC flows upward, so the current is in the direction **A → B → C → D** (i.e. C → D in the right side). Reverse rotation reverses the direction.
2024 Q8(e)	Transformer, 2000 turns input, 100 turns output, output 10 V, input current 0.2 A, 100 % efficient. Supply voltage?	$V_p/V_s = N_p/N_s \Rightarrow V_p = 10 \times 2000 / 100 =$ **200 V**. (Check: power in = 200 × 0.2 = 40 W = 10 V × 4 A.)
2025 Q4(c)	Why are transformers not 100 % efficient?	Some energy is lost: **heat in the copper coils (I^2R loss), eddy currents in the iron core, hysteresis loss, flux leakage (not all flux links the secondary) and sound/vibration**.

2025 Q5(d)	Copper rod XY in a magnetic field (field lines drawn from right to left) joined to a battery: show the direction of current and force.	Redraw the figure. Taking the cell's longer line as the positive terminal, the current flows from the **positive terminal to X, along the rod from X to Y** and back to the negative terminal (if the cell is connected the other way, current and force reverse). By Fleming's left-hand rule: first finger = field (to the left), second finger = current (X → Y, down the rod), thumb = **force on the rod, into the page**.
2025 Q8(e)	Why use a step-up transformer to carry electricity from a hydropower station to far places?	The power P = V I stays the same, so raising the voltage lowers the current. A **smaller current means much less heat loss (I^2R) in the long transmission wires**, so more power reaches the distant users; step-down transformers then reduce the voltage for safe use.

Chapter 8: Waves, the Electromagnetic Spectrum & Communication

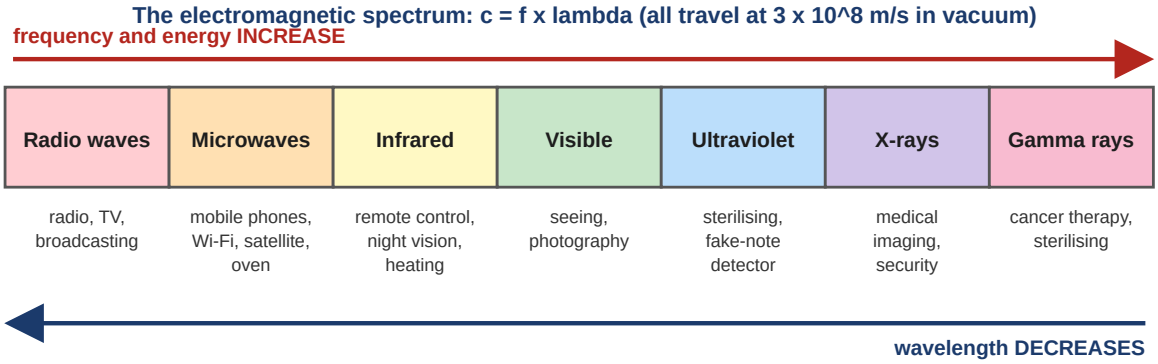
The most question-rich chapter: 10–12 items in every paper

8.1 Must-know summary

Term	Meaning / rule
Transverse vs longitudinal	**Transverse:** vibration **perpendicular** to the direction of travel (light, all EM waves, X-ray, radio, water ripples). **Longitudinal:** vibration **parallel** to the direction of travel, made of compressions and rarefactions (sound). Sound needs a medium; EM waves do not.
Wave equation	$v = f\lambda$. For EM waves $c = 3 \times 10^8$ m/s in a vacuum, so **frequency ∝ 1/wavelength**: when frequency increases, wavelength decreases.
EM waves (common properties)	Transverse; travel in a vacuum at the speed of light; carry energy; can be reflected, refracted, diffracted; uncharged (not deflected by fields).
Diffraction	Waves bend around obstacles or spread through openings, most when the **wavelength is comparable to or larger than the obstacle**. Radio waves (long λ) bend around hills; sound is heard around corners and from the next room.
Analogue vs digital	**Analogue:** continuous values (sine wave), used in radio/TV broadcasting; affected by noise. **Digital:** only 0 and 1 (square wave); noise-free regeneration, easy storage and processing, secure, high bandwidth; used in computers, mobile phones, blood-pressure monitors.
Modulation	Putting the information (input signal) on a carrier wave. **AM:** the **amplitude** of the carrier varies with the signal. **FM:** the **frequency** varies and the amplitude is constant.
Propagation of radio waves	**Ground waves** (low frequency) follow the Earth's surface; **sky waves** (short-wave) are **reflected by the ionosphere** to far places; **space waves** (TV, FM, microwaves) pass through the ionosphere (satellite links) or travel in a **straight line of sight** between towers.
Wireless technologies	Bluetooth (about 10 m), Wi-Fi (microwaves), mobile networks (microwaves), infrared remote controls (line of sight), satellite links (microwaves).
Wired: optical fibre vs copper	Fibre carries light by total internal reflection: **very low signal loss, huge bandwidth/data, no electromagnetic interference**, light, secure, does not corrode. Copper loses signal and picks up interference.

Wave	Main uses	Harm / precaution
Radio	radio and TV broadcasting, mobile communication, radio telescopes	very low energy; harmless
Microwaves	mobile phones, Wi-Fi, satellite and radar links, microwave ovens	heat body tissue; limit phone use, keep transmitters away from homes
Infrared	TV remote, night-vision and thermal photography (not scattered easily; objects emit it), heating, greenhouse effect (keeps Earth warm), medical therapy	burns at high intensity
Visible	vision, photography, optical fibre	—
Ultraviolet	sterilising, **counterfeit-note detector** (fluorescence), treating neonatal jaundice (phototherapy), vitamin D	skin burns/cancer, eye damage: sunscreen, goggles
X-rays	medical imaging (bones, strokes, tumours), airport security	ionising: avoid unnecessary exposure; lead aprons; harmful to a foetus

Gamma	killing cancer cells, sterilising equipment, medical tracers (iodine-123)	ionising, most penetrating: lead shielding, limit time and keep distance



RADIO (longest wavelength, ~1 m - km) -> GAMMA (shortest wavelength, < 10^-11 m)

Fig 8.1 The electromagnetic spectrum (2025 Q2d)

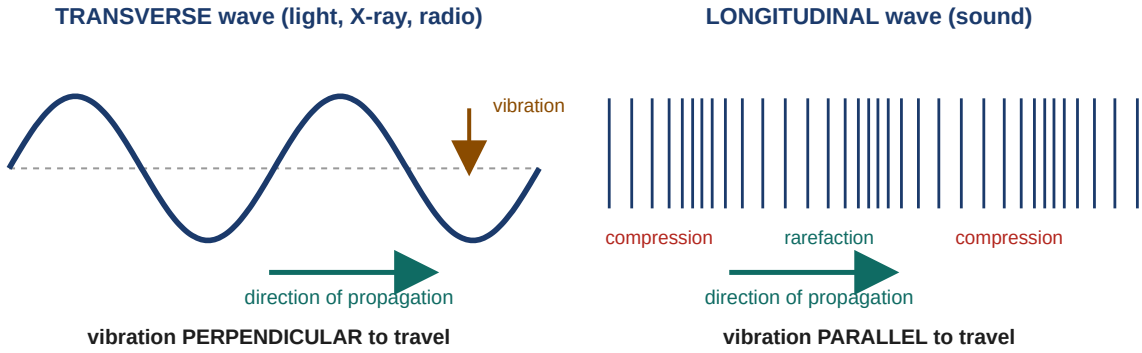


Fig 8.2 Transverse and longitudinal waves (2025 Q7e)

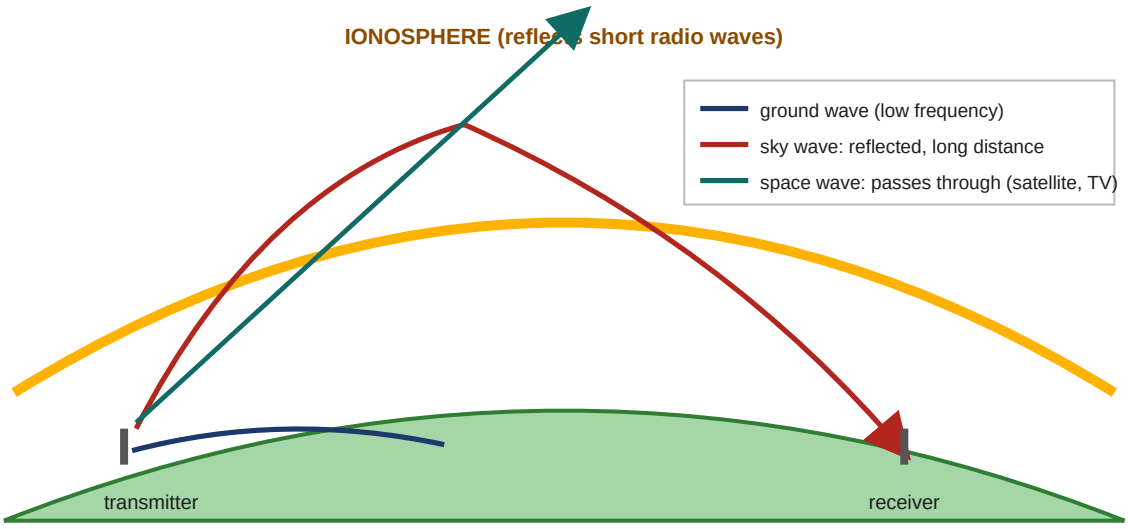


Fig 8.3 Ground, sky and space waves (2024 Q3b)

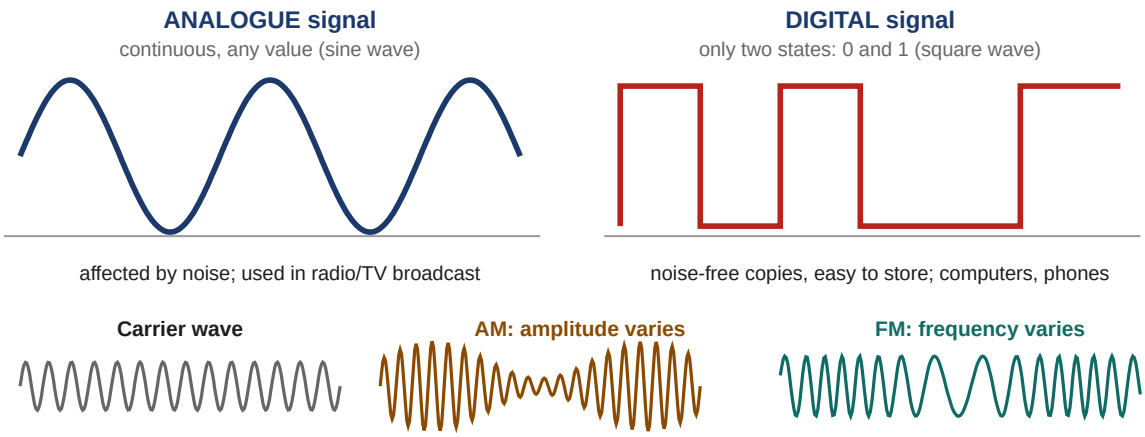


Fig 8.4 Analogue and digital signals; AM and FM (2023 Q3e)

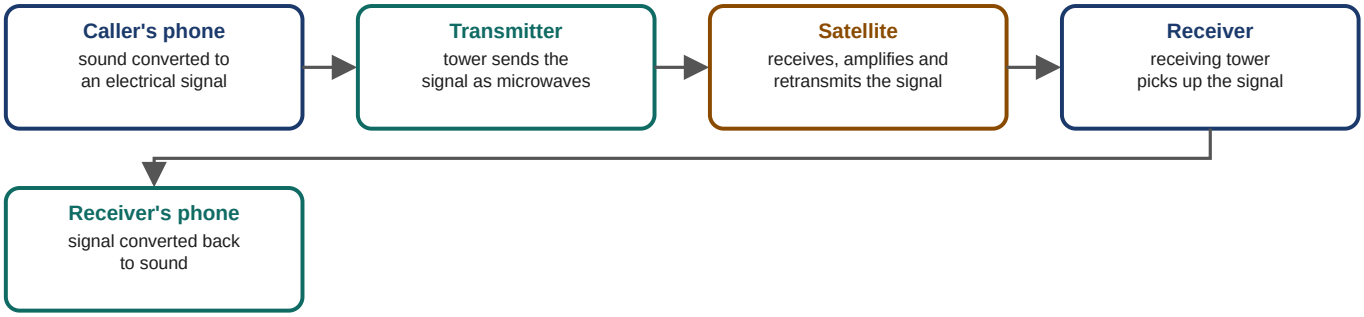


Fig 8.5 A phone call over a long distance (2023 Q6d)

8.2 Solved questions — Section A

Ref	Question	Ans	Why
2023 MCQ-ix	Infrared radiation is used for photographs at night because the wave…	**D**	Infrared (emitted by warm objects) is **not scattered easily** by haze or dark air, so thermal photographs are possible. (Option C, heating effect, is also a property of IR, but D explains the photography.)
2023 MCQ-x	A teacher hears students in the next classroom. Property of sound?	**C**	Diffraction — sound bends around the door opening and corners.
2023 MCQ-xx	Increasing order of frequency for W 2×10^{-5}, X 2×10^{-3}, Y 5×10^{-5}, Z 5×10^{-3} m	**B**	$f \propto 1/\lambda$: the longest wavelength has the lowest frequency: **Z < X < Y < W**.
2023 MCQ-xxi	Digital signal: I uses 0 and 1; II bandwidth is high; III denoted by sine waves	**A (I and II)**	I and II are correct; III is false (sine waves are analogue; digital signals are square waves).
2024 MCQ-iv	Wave used to destroy cancer cells	**B**	Gamma rays.
2024 MCQ-vi	Technology using microwaves	**A**	Wi-Fi.
2024 MCQ-xiii	Use of X-rays in biomedical engineering	**B**	Detecting strokes, tumours and fractures (imaging).
2024 MCQ-xvi	Digital signal of a blood-pressure monitor	**A (I)**	A digital signal has only two levels, 0 and 1 (square pulses between 0 and 1).
2024 MCQ-xviii	Application that uses analogue signals	**C**	Television and radio broadcasting.

2024 MCQ-xxiii	Wave for a space telescope (longer wavelengths penetrate gas and dust better)	**C (radio waves)**	Radio waves have the longest wavelength of the choices, so they are scattered and absorbed least by dust. (Infrared is the next best and is also used by space telescopes.)
2025 MCQ-x	Wave that keeps the Earth warm through the greenhouse effect	**C**	Infrared rays.
2025 MCQ-xi	Tenzing hears her phone ringing from behind the building. Why?	**D**	Sound waves diffract around obstacles and through openings.
2025 MCQ-xxiii	Satellite, TV remote, sterilising hospital equipment	**B (row 2)**	Microwaves, infrared radiation, ultraviolet rays. (X-rays are not used for sterilising.)
2025 MCQ-xxiv	Blocked TV remote path shows…	**D**	Infrared radiation travels along a straight line.

Ref	Question	Answer and working
2023 Fill-v	Wireless communication used to transfer data within 10 metres is called…	**Bluetooth**
2023 Match-v	Waves with minimum obstruction by large structures	**f. radio waves** (long wavelength, large diffraction).
2025 Fill-v	Radio waves reach far places for broadcasting because of their…	**long wavelength** (so they diffract around obstacles and can be reflected by the ionosphere).
2025 T/F-iv	Ultraviolet rays are used in treating neonatal jaundice in infants.	**TRUE** (as taught in the syllabus; in hospitals the phototherapy light is close to the UV end, blue light).
2025 Match-iii	Satellite communication	**a. micro-waves**

8.3 Solved questions — Section B

Ref	Question	Answer and working
2023 Q2(c)(i)	A radiation X is focused on the bulb of a clinical thermometer and the mercury rises rapidly. Name X.	**Infrared radiation** (heat radiation).
2023 Q2(c)(ii)	ONE property demonstrated.	It has a **heating effect** (and it can be focused/refracted by a lens like light).
2023 Q2(d)	Why is a digital signal preferred to an analogue signal?	A digital signal **is less affected by noise and can be regenerated without loss**, can carry more information, is easy to store, process and encrypt (secure).
2023 Q3(e)	Waves A and B are modulated. Which is amplitude modulated?	**Wave A**: its **amplitude varies** in step with the input signal while its frequency stays constant. In wave B the frequency changes with a constant amplitude (frequency modulation).
2023 Q5(f)	TWO similarities between visible light and radio waves.	Both are **electromagnetic waves** and **transverse**; both travel at 3×10^8 m/s in a vacuum and need no medium (they also carry energy and can be reflected, refracted and diffracted).
2023 Q6(d)	Explain long-distance communication with the key words (caller's phone, transmitter, satellites, receiver, receiver's phone) and identify the wave.	See Fig 8.5: the **caller's phone** changes sound into an electrical signal → the **transmitter** (tower) sends it as radio/micro waves to a **satellite** → the satellite amplifies and retransmits it to the **receiver** (tower) → the **receiver's phone** changes the signal back into sound. The wave used: **microwaves** (radio waves).

2023 Q7(e)	Compare and contrast gamma rays and hard X-rays.	**Differences:** gamma rays come from the **nucleus** (radioactive decay), X-rays are produced **outside the nucleus** (fast electrons striking a metal target); gamma rays generally have higher energy/frequency and penetrate more. **Similarities:** both are EM waves, transverse, travel at c, uncharged, ionising and harmful to living cells.
2023 Q9(e)	Why are pregnant women advised not to take X-rays unless required?	X-rays are **ionising radiation** that can damage the cells and DNA of the rapidly growing foetus, causing mutations, birth defects or cancer.
2023 Q9(f)	Optical fibre is better than copper cable. Agree? One reason.	**Agree**: an optical fibre loses very little signal over long distances and carries much more information (larger bandwidth); it is also free from electromagnetic interference.
2024 Q3(b)	Overview of waves X and Y; why Y for BBS radio?	**Wave X** passes **through** the ionosphere into space (a space wave, e.g. satellite/TV/microwaves; travels in a straight line). **Wave Y** is **reflected back by the ionosphere** to the Earth (a sky wave, short-wave radio). BBS uses **Y** because the reflected waves can reach far, mountainous places beyond the horizon where direct line-of-sight is blocked.
2024 Q5(e)	TWO differences between radio waves and gamma rays.	**Radio waves:** longest wavelength, lowest frequency and energy; produced by oscillating charges in aerials; harmless, used in communication. **Gamma rays:** shortest wavelength, highest frequency and energy; emitted by unstable nuclei; ionising and harmful, very penetrating.
2024 Q7(b)	Should we be cautious about cell phone radiation?	**Yes, it is sensible to be careful.** Cell phones use microwaves, which are non-ionising but can heat tissue; the long-term effect of prolonged exposure is not fully known and many people (children especially) use phones for long periods. Precautions: limit use, use hands-free/speaker, keep the phone away from the body and do not sleep next to it.
2024 Q7(d)	Why are cell phone transmitters on hill tops?	Microwaves **travel in straight lines (line of sight) and have short wavelengths that diffract very little**, so hills and buildings block them. A high position clears the obstacles and covers a wider area.
2024 Q8(a)	Which wave does a counterfeit-money detector use? Why?	**Ultraviolet rays**: they make the special inks and security threads on real notes **fluoresce** (glow), which is invisible in ordinary light.
2024 Q9(e)	How do radio waves help communication when a mountain lies between the transmitter and receiver?	Radio waves have a **long wavelength, so they diffract around the mountain** and can also be reflected by the ionosphere, reaching the receiver behind the mountain where microwaves or light would be blocked.
2025 Q2(d)	How is the EM spectrum organised? What happens to wavelength as frequency increases?	It is arranged by **increasing frequency (and energy)**, i.e. decreasing wavelength: radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, gamma rays (Fig 8.1). As the frequency increases, the **wavelength decreases** ($c = f\lambda$ with c constant).
2025 Q3(d)	How does diffraction of radio waves affect propagation over long distances and around obstacles?	Radio waves have long wavelengths, comparable to hills and buildings, so they **bend (diffract) around obstacles and along the curve of the Earth**, so signals reach places behind hills and far away (sky waves reflected by the ionosphere reach even farther).
2025 Q6(e)	Cell phone: type of signal (1) between the satellite and tower, (2) between the mobile and the person.	(1) **digital** (square pulses); (2) **analogue** (the phone changes the digital signal back into the analogue sound wave for the listener).
2025 Q7(e)(i)	Which row shows the nature of light, sound and X-rays?	**Row 2**: light transverse, sound longitudinal, X-rays transverse.

2025 Q7(e)(ii)	How do you distinguish longitudinal and transverse waves?	By the **direction of vibration compared with the direction of travel**: in transverse waves the particles vibrate perpendicular to it (crests and troughs); in longitudinal waves they vibrate parallel to it (compressions and rarefactions). Also, longitudinal waves (sound) need a medium (Fig 8.2).
2025 Q8(d)	Minimise prolonged exposure to EM waves A and B (A is between gamma and ultraviolet; B between infrared and radio).	**A = X-rays:** avoid unnecessary scans, wear lead aprons, keep the exposure time short and stay behind shielding. **B = microwaves:** do not stay near transmitters, use phones with hands-free and for short times, keep damaged microwave ovens away/repaired.
2025 Q9(f)	Copper or optical fibre for a weak TV signal? TWO reasons.	**Optical fibre**: (1) very **low signal loss** (attenuation) over long distances so the signal stays strong; (2) **large bandwidth** carries many channels with good quality; also free from interference and secure.

Chapter 9: Radioactivity & Nuclear Energy

A decay equation and an alpha/beta/gamma question appear every year

9.1 Must-know summary

Term	Meaning / rule
Nucleus notation	${}^{A}_{Z}X$: A = **mass (nucleon) number** = protons + neutrons; Z = **atomic (proton) number**. **Isotopes** have the same Z but different A.
Radioactivity	The spontaneous emission of radiation (α, β, γ) from **unstable nuclei**, which lose particles/energy to become stable (they do not "receive" extra protons or neutrons).
Alpha (α)	Helium nucleus ${}^{4}_{2}He$ (2 protons + 2 neutrons), charge +2, heavy. **Highest ionising power, lowest penetration** (stopped by paper/skin). Bent towards the negative plate.
Beta (β)	A fast electron ${}^{0}_{-1}e$ (from a neutron changing to a proton), charge −1, light. **Medium** ionising and penetration (stopped by a few mm of aluminium). Bent strongly towards the positive plate.
Gamma (γ)	High-energy EM wave, no charge or mass. **Lowest ionising power, highest penetration** (reduced only by thick lead/concrete). Not deflected by electric or magnetic fields.
Decay equations	**Alpha:** A decreases by 4, Z by 2. **Beta:** A unchanged, Z increases by 1. **Gamma:** A and Z unchanged. Totals of A and of Z must balance on both sides.
Uses	Treating cancer (gamma), sterilising equipment, medical tracers (iodine-123 for the thyroid: gamma passes out through the body and has low ionising power), smoke detectors, tracers in industry.
Hazards & safety	Ionising radiation damages cells (burns, cancer, mutations). **Precautions:** handle with long tongs, store in lead-lined containers, wear protective clothing and a radiation badge (dosimeter), keep a distance and minimise the exposure time, no eating or drinking in the area, proper disposal of waste.
Nuclear energy	Energy released by nuclear **fission** (heavy nuclei such as uranium split, in reactors) or **fusion** (light nuclei join, in stars). It is **not the same as radioactivity**, which is the spontaneous decay of unstable nuclei (uncontrolled and not used to generate power directly).

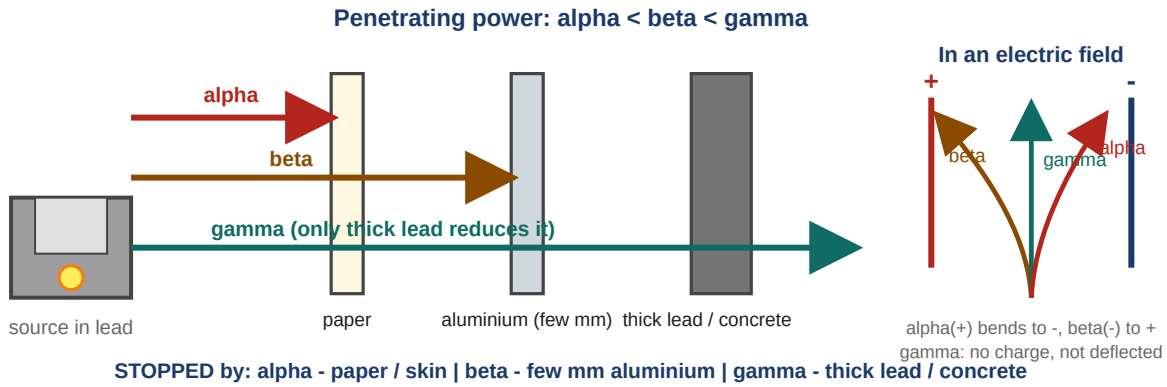


Fig 9.1 Penetrating power of alpha, beta and gamma and their deflection in an electric field

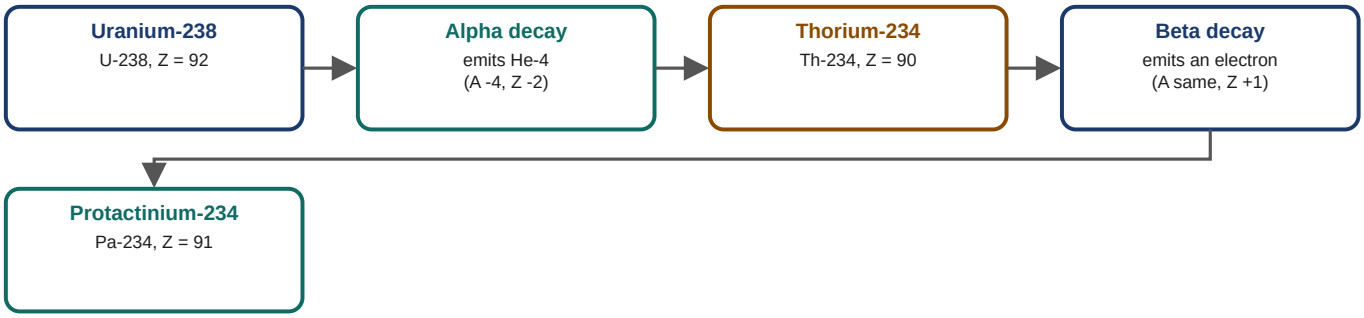


Fig 9.2 Decay chain of uranium-238 (2025 Q4d)

9.2 Solved questions — Section A

Ref	Question	Ans	Why
2023 MCQ-xi	$^{244}_{94}X \rightarrow {}^{240}_{Z}Y + \alpha$: find Z	**B**	Alpha removes 2 protons: Z = 94 − 2 = **92** (and A = 244 − 4 = 240).
2023 MCQ-xxii	Atom A has the highest ionising and lowest penetrating power; atom B the opposite. Which atom and which ray?	**B**	Highest ionising and lowest penetrating = **alpha**: Atom A and alpha rays.
2024 MCQ-vii	X(A,Z) → Y(A,Z+1) → B(A−4,Z−2) → B(A−4,Z−2): particles emitted in sequence	**D**	First step: A unchanged, Z + 1 = **β**. Second: A − 4, Z − 2 = **α**. Third: no change = **γ**. Sequence **β, α, γ**.
2024 MCQ-xiv	Symbol of alpha decay	**C**	$^{4}_{2}He$ (III): the helium nucleus.
2025 MCQ-xii	Highest penetrating power	**D**	Gamma radiation.
2025 MCQ-xxv	Why gamma (not alpha or beta) for examining the thyroid?	**C**	Gamma radiation passes through the body so it can be detected outside by the detector (alpha and beta are absorbed).

Ref	Question	Answer and working
2023 T/F-iv	The mass number of an element remains the same in beta decay since the number of protons increases and neutrons decrease.	**TRUE** (a neutron changes into a proton: Z + 1, neutrons − 1, A unchanged).
2024 Fill-iv	$^{241}_{95}Am \rightarrow {}^{237}_{93}Np +$ _____	**$^{4}_{2}$He (an alpha particle)** — A: 241 − 237 = 4; Z: 95 − 93 = 2.
2025 T/F-v	Atoms of unstable nuclei attain stability by receiving extra protons or neutrons.	**FALSE** — they become stable by *emitting* particles or energy (α, β, γ).

9.3 Solved questions — Section B

Ref	Question	Answer and working
2023 Q2(e)	TWO safety precautions when handling a radioactive substance.	(1) Handle it with **long tongs and keep it in a lead-lined container** (distance and shielding). (2) Wear **protective clothing/gloves and a radiation badge**, and keep the exposure time short.
2023 Q4(e)(i)	P (mass number 226, atomic number 88) undergoes alpha decay then two beta decays. Nuclide notation of S.	Alpha: A = 226 − 4 = 222, Z = 88 − 2 = 86. Each beta: Z + 1, A unchanged: after two betas Z = 88, A = 222. So S = **$^{222}_{88}$S**.

2023 Q4(e)(ii)	Compare the atomic and mass numbers of P and S.	**Atomic number: the same (88)** — P and S are **isotopes**. **Mass number: S (222) is 4 less than P (226).**
2023 Q5(b)	11 % of the world's electricity is nuclear. Should Bhutan generate nuclear energy?	**No.** Bhutan already has abundant, clean and cheap hydropower. A nuclear plant needs huge investment and advanced technology, imported uranium, safe disposal of radioactive waste for thousands of years, and carries the risk of accidents (Chernobyl, Fukushima), especially in an earthquake-prone Himalayan country with a small grid.
2023 Q6(e)(i)	Diagram: a source in a lead box; rays deflected between + and − plates. Name radiations P and R.	P bends towards the **positive plate**, so it is negative: **beta (β)**. R bends towards the **negative plate**: **alpha (α)**. (Q is undeflected: gamma.)
2023 Q6(e)(ii)	Why is ray Q not deflected?	Q is **gamma radiation, an electromagnetic wave with no charge**, so electric and magnetic fields do not deflect it.
2023 Q8(f)	Should schools use radioactive materials in laboratories?	**No, not without strict control.** Radiation ionises and damages living cells (burns, cancer, mutations); students may mishandle, lose or be exposed to the source, and storage and disposal need special facilities. Safer ways: simulations, videos and demonstrations by trained experts using very weak sealed sources.
2024 Q5(d)	TWO safety precautions in a nuclear power plant.	(1) **Shield the reactor with thick concrete and lead** and use remote handling; (2) **wear protective suits with radiation badges, limit the exposure time and monitor radiation levels**; also safe storage of waste and regular training.
2024 Q8(b)	Which characteristic (mass or charge) determines the ionising power and which the penetration depth of α, β, γ?	**Charge determines the ionising power** (alpha +2 is highest, beta −1 medium, gamma 0 lowest). **Mass determines the penetration** (heavy alpha is stopped by paper, light beta by aluminium, massless gamma penetrates most).
2024 Q9(a)	$^{14}_{7}N + {}^{4}_{2}\alpha \rightarrow X + {}^{1}_{1}H$. Proton number and nucleon number of X?	Nucleon number: 14 + 4 = 18 = X + 1 ⇒ **17**. Proton number: 7 + 2 = 9 = X + 1 ⇒ **8**. X is $^{17}_{8}O$ (oxygen-17).
2025 Q4(d)	U-238 (Z = 92) alpha decays to thorium, then thorium beta decays to protactinium. Write the equations.	$^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$ and $^{234}_{90}Th \rightarrow {}^{234}_{91}Pa + {}^{0}_{-1}e$ (Fig 9.2).
2025 Q5(e)	Is nuclear energy the same as radioactivity?	**No.** Radioactivity is the spontaneous decay of unstable nuclei emitting α, β or γ (it cannot be controlled). Nuclear energy is the energy released in nuclear reactions (fission or fusion), controlled in reactors to produce electricity. A reactor uses radioactive material (uranium) but the two ideas are different.
2025 Q6(f)	The Japanese PM ate fish from Fukushima after treated water was released. What conclusions about the effects of nuclear radiation on living things?	Radiation is **harmful in large doses** (it damages cells, causes cancer and mutations, and radioactive substances can build up in the food chain), which is why the water was treated and monitored. The PM's action suggests that **when the radioactivity is reduced below the safety limit, the harm is negligible**: the effect depends on the dose and the type of radiation, so continuous monitoring is needed.

Chapter 10: Space Exploration & the Universe

An "explain / compare / justify" question comes from this chapter every year

10.1 Must-know summary

Topic	Key points
Orbiter spacecraft	Enters and stays in **orbit** around a planet or moon; studies it for a **long time**, maps it and sends data (needs fuel to slow down and be captured). E.g. Mars orbiters, Chandrayaan orbiter.
Flyby mission	Passes a planet **once without orbiting or landing**, collecting data briefly; needs little fuel, can visit several bodies. E.g. **Voyager 1** (Jupiter, Saturn), New Horizons.
Rover / lander	Lands and **moves on the surface**, collects soil and rock samples and searches for signs of life (Perseverance, Curiosity). Cheaper and safer than humans, but slow and limited.
Human space exploration	Astronauts travel (Apollo 11, 1969): flexible and can make decisions and bring samples back, but **very expensive** and risky to life. Its main impact: **technological advances with wide uses on Earth** (satellites, medical devices, materials).
Robotic exploration	Use of **autonomous or remote-controlled spacecraft** to study celestial bodies.
Pros and cons	**Advantages:** more knowledge of the universe, minerals and resources, new technology, inspires young scientists. **Disadvantages:** very expensive (economy diverted, possibly leading to poverty), risk to astronauts, space debris.
Life on other planets	Needs **liquid water, suitable temperature, an atmosphere and an energy source** (conditions we know from Earth). Habitable-zone planets, water on Mars and Europa suggest possibility, but no life has been found yet.
Big Bang theory	The universe began about 13.8 billion years ago from an extremely hot, dense state and has been **expanding**. Evidence: galaxies moving apart (red shift) and the cosmic microwave background. Critics point out that it does not explain what existed before or caused the bang, the singularity cannot be tested, and observations such as very old structures or dark matter/energy are not fully explained.
Red shift	Light from receding galaxies has a **longer wavelength** (shifted towards red): evidence that the universe is **expanding**. (A blue shift would mean the source is approaching.)
Stars	Stars shine because of **nuclear fusion** of light elements (hydrogen → helium) in their cores. A massive star ends as a **supernova** (a huge explosion that can outshine a galaxy) leaving a neutron star or black hole.
Origin of the solar system	Nebular hypothesis (Fig 10.1): a cloud of gas and dust collapses, flattens into a spinning disc, forms the Sun and, by accretion, planets.

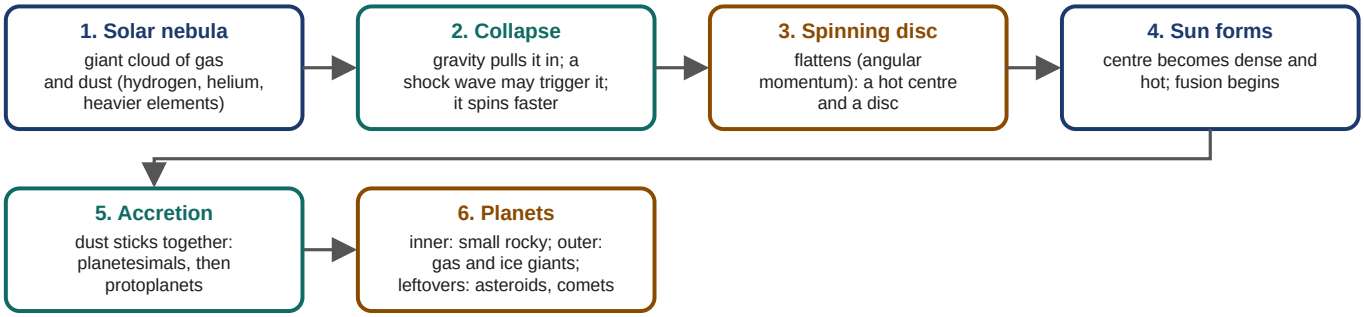


Fig 10.1 Model for the formation of the solar system (2025 Q2e)

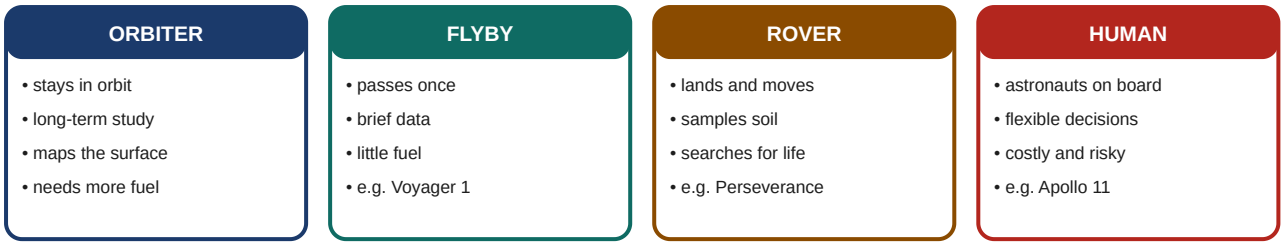


Fig 10.2 Types of space missions

10.2 Solved questions — Section A

Ref	Question	Ans	Why
2023 MCQ-xii	Collect soil samples and look for signs of life on Mars: which mission?	**A**	A rover lands and moves on the surface to collect samples.
2023 MCQ-xxiv	Disadvantages of space exploration (I economy diverted leading to poverty, II minerals, III knowledge, IV expensive, V risk to astronauts)	**B (I, IV and V)**	II and III are advantages.
2024 MCQ-xvii	Voyager 1 (Jupiter and Saturn, longest distance) is an exemplar of…	**B**	a **flyby** space mission.
2024 MCQ-xxiv	Robotic exploration of space	**C**	Use of autonomous spacecraft to study and explore celestial bodies.
2025 MCQ-xiii	Primary impact of human space exploration	**D**	It has led to technological advances with wide-ranging applications on Earth.

Ref	Question	Answer and working
2023 T/F-v	The expansion of the universe is evident from cosmological blue shift.	**FALSE** — it is shown by the cosmological **red shift**.
2024 T/F-ii	Energy is produced in the cores of main-sequence stars when lighter elements fuse into heavier elements.	**TRUE**
2025 Fill-i	Explosions caused by the collapse of massive stars under their own gravity, which can outshine a galaxy, are called…	**supernova**
2025 Match-v	Every matter in the universe moves away from the observer	**f. cosmological redshift**

10.3 Solved questions — Section B

Ref	Question	Answer and working
2023 Q2(f)	Many cosmologists believe in a big bang but some oppose it. Possible reasons for disagreement?	(1) The theory **cannot explain what existed before the bang or what caused it**; the starting point (singularity) cannot be tested by experiment. (2) Some observations (e.g. very old galaxies, dark matter and dark energy) or alternative models (such as the steady-state model) are not fully explained by it.
2023 Q4(f)	Differentiate orbiter and flyby space exploration.	**Orbiter:** goes into **orbit around the planet** and studies it over a long time; gives detailed data; needs more fuel and cost. **Flyby:** **passes the planet only once** without orbiting or landing; gets data only for a short time, but needs little fuel and can visit several planets (Voyager 1).

2023 Q7(f)	Is there a possibility of life on other planets?	**Yes, it is possible.** The universe has billions of galaxies and planets; some lie in the habitable zone where liquid water can exist; signs of water have been found on Mars and Europa. Life needs water, a suitable temperature, an atmosphere and energy, so such planets may support life, although none has been confirmed yet.
2024 Q5(c)	Why examine the conditions necessary for life on the Earth to study life in outer space?	The Earth is the **only place where life is known to exist**, so its conditions (liquid water, suitable temperature, an atmosphere, an energy source) are the reference for deciding which planets might support life.
2025 Q2(e)	Create a theoretical model for the formation of the solar system, with mechanisms and an account of the planet types.	See Fig 10.1. **Model:** a giant cloud of gas and dust (solar nebula) **collapses under gravity** (perhaps triggered by a nearby supernova) and **spins faster and flattens into a disc**. The centre becomes the **Sun** when fusion starts. Dust grains **collide and stick (accretion)** into planetesimals and then planets. **Inner planets** are small and rocky because only metals and rock could condense in the heat; **outer planets** are large gas/ice giants because they formed in the cold where ices and gases could gather. The model explains why the planets orbit in the same direction and plane, the asteroid belt and comets (leftover material).
2025 Q3(e)	Apollo 11 (crewed lunar landing) vs Mars 2020 Perseverance (rover): advantages and limitations.	**Apollo 11:** *advantages* — humans decide, adapt and perform complex tasks, and brought samples back; *limitations* — extremely costly, needs life-support, risks astronauts' lives, short stay. **Perseverance rover:** *advantages* — cheaper, no risk to human life, can work for years in harsh conditions and reach distant Mars; *limitations* — slow, limited instruments, signal delay so it cannot react quickly, cannot bring samples back by itself and may break down.

Formula Sheet & Last-Minute Revision

Physics — everything on two pages

Formulas

Topic	Formulas	Units
Moments	Moment = F × d; clockwise = anticlockwise; couple = F × (distance between forces)	N m
Gravitation	$F = G\,m_1 m_2 / r^2$ ($G = 6.67 \times 10^{-11}$); W = m g; v = g t	N; N; m/s
Work / energy / power	W = F d; KE = ½ m v²; GPE = m g h; P = W/t = F v; efficiency = output/input × 100 %	J; J; J; W; %
Pressure	P = F/A; P = ρ g h; $F_1/A_1 = F_2/A_2$	Pa (N/m²)
Heat	Q = m c ΔT; Q = m L; heat lost = heat gained (calorimetry)	J
Electricity	V = I R; P = V I = I²R = V²/R; H = I² R t; I = Q/t	V; W; J; A
Transformer	$V_s/V_p = N_s/N_p$; $V_p I_p = V_s I_s$	V; A
Waves	v = f λ; c = 3×10^8 m/s	m/s
Radioactivity	α: A − 4, Z − 2; β: A same, Z + 1; γ: no change	—

Fleming's rules and key facts

Item	Remember
Left hand (motor)	Thumb = Force; First finger = Field; Second finger = Current.
Right hand (generator)	Thumb = Motion; First finger = Field; Second finger = Induced current.
Stability	Low C.G. + wide base = stable; topples when the line of gravity leaves the base.
Terminal velocity	Drag = weight, net force = 0, acceleration = 0, constant velocity.
Calculations	Write: formula → substitution → answer with unit (marks are given for each step). Use g = 10 m/s² unless told otherwise.
Justify / design questions	Give a clear decision (Yes/No) and TWO reasons linked to physics ideas from the chapter.

Common mistakes

• Forgetting units (N m, Pa, W, J). • Using grams instead of kilograms in Q = mcΔT (check the unit of c). • Mixing up Fleming's left and right hands. • Saying "radioactivity = nuclear energy". • Writing "beta has no mass" or "gamma is a particle". • Saying a bigger wire has more resistance. • Circling two options in an MCQ.